خدمتكم عبادة نتقرب بها الى الله

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دفتر

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تفاضل وتكامل 2



Calculus II

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بقدر الأكد تكتب العالي

ومن طلب العلا سهر الليالي

ومن رام العلا من غير كد

أضاع العمر في طلب المحال

تروم العز ثم تنام ليلاً

يغوص البحر من طلب اللآلي

الشامخي

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Integration formulas

① $\int a\,dx = ax + c$

② $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad , n \neq -1$

③ $\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \quad , n \neq -1$

④ $\int x^{-1}\,dx = \int \frac{1}{x}\,dx = \ln|x| + c$

⑤ $\int \frac{dx}{ax+b} = \frac{1}{a}\ln|ax+b| + c$

⑥ $\int \frac{f'}{f}\,dx = \ln|f| + c$.

⑦ $\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + c$. or $\int k^{ax+b}\,dx = \frac{k^{ax+b}}{a \ln k} + c$

⑧ $\int \sin(ax+b)\,dx = -\frac{\cos(ax+b)}{a} + c$.

⑨ $\int \cos(ax+b)\,dx = \frac{\sin(ax+b)}{a} + c$

⑩ $\int \sec^2(ax+b)\,dx = \frac{\tan(ax+b)}{a} + c$.

⑪ $\int \csc^2(ax+b)\,dx = -\frac{\cot(ax+b)}{a} + c$.

⑫ $\int \sec(ax+b)\tan(ax+b)\,dx = \frac{\sec(ax+b)}{a} + c$.

⑬ $\int \csc(ax+b)\cot(ax+b)\,dx = -\frac{\csc(ax+b)}{a} + c$.

⑭ $\int \tan(ax+b)\,dx = \frac{-1}{a}\ln|\cos(ax+b)| + c$.

⑮ $\int \cot(ax+b)\,dx = \frac{1}{a}\ln|\sin(ax+b)| + c$

(16) $\int \sinh(ax+b)\,dx = \cosh(ax+b) + c$

(17) $\int \cosh(ax+b)\,dx = \sinh(ax+b) + c.$

(18) $\int \operatorname{sech}^2(ax+b)\,dx = -\tanh(ax+b) + c.$

(19) $\int \operatorname{csch}^2(ax+b)\,dx = -\coth(ax+b) + c.$

(20) $\int \operatorname{sech}(ax+b)\tanh(ax+b)\,dx = -\operatorname{sech}(ax+b) + c.$

(21) $\int \operatorname{csch}(ax+b)\coth(ax+b)\,dx = -\operatorname{csch}(ax+b) + c.$

(22) $\int \frac{dx}{\sqrt{a^2-b^2x^2}} = \frac{1}{b}\sin^{-1}\frac{bx}{a} + c.$

(23) $\int \frac{dx}{a^2+b^2x^2} = \frac{1}{ab}\tan^{-1}\frac{bx}{a} + c.$

(24) $\int \frac{dx}{x\sqrt{b^2x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{b}{a}x + c.$

or use

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c$$

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c.$$

$$\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{x}{a} + c.$$

Hyperbolic functions

$$e^x = \underbrace{\frac{e^x + e^{-x}}{2}}_{\text{even f=}} + \underbrace{\frac{e^x - e^{-x}}{2}}_{\text{odd f=}}$$

Define $\sinh x = \frac{e^x - e^{-x}}{2}$, $D = R : (-\infty, \infty)$

$\cosh x = \frac{e^x + e^{-x}}{2}$, $D : (-\infty, \infty)$, $R = [1, \infty)$

$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, $D : (-\infty, \infty)$, $R : (-1, 1)$

$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$, $D : (-\infty, \infty) \setminus \{0\}$, $R : (-\infty, -1) \cup (1, \infty)$

$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$, $D : (-\infty, \infty)$, $R : (0, 1]$

$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$, $D = R = (-\infty, \infty) \setminus \{0\}$

* Identities :

$\cosh x + \sinh x = e^x$

$\cosh x - \sinh x = e^{-x}$

$\cosh^2 x - \sinh^2 x = 1$

$1 - \tanh^2 x = \operatorname{sech}^2 x$

$\coth^2 x - 1 = \operatorname{csch}^2 x$

$\sinh(2x) = 2 \sinh x \cosh x$

$\cosh(2x) = \cosh^2 x + \sinh^2 x$

$= 2 \sinh^2 x + 1$

$= 2 \cosh^2 x - 1$

$\sinh(x \pm y) = \sinh x \cosh y \pm \sinh y \cosh x$

$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$

$\sinh(-x) = -\sinh x$

$\cosh(-x) = \cosh x$

Derivatives

f	f'
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$
$\operatorname{sech} x$	$-\operatorname{sech} x \tanh x$
$\operatorname{csch} x$	$-\operatorname{csch} x \coth x$
$\coth x$	$-\operatorname{csch}^2 x$

Integration

$$\int \sinh x\,dx = \cosh x + C$$
$$\int \cosh x\,dx = \sinh x + C$$
$$\int \operatorname{sech}^2 x\,dx = \tanh x + C$$
$$\int \operatorname{csch}^2 x\,dx = -\coth x + C$$
$$\int \operatorname{sech} x \tanh x\,dx = -\operatorname{sech} x + C$$
$$\int \operatorname{csch} x \coth x\,dx = -\operatorname{csch} x + C$$

Examples:

Find ① $\sinh 3 = \dfrac{e^3 - e^{-3}}{2}$ ② $\cosh 0 = \dfrac{e^0 + e^{-0}}{2} = 1$

③ $\tanh(\ln 4) = \dfrac{e^{\ln 4} - e^{-\ln 4}}{e^{\ln 4} + e^{-\ln 4}} = \dfrac{4 - \frac{1}{4}}{4 + \frac{1}{4}} = \dfrac{15}{17}$

④ $\operatorname{sech}(-\ln 2) = \dfrac{2}{e^{-\ln 2} + e^{\ln 2}} = \dfrac{2}{\frac{1}{2} + 2} = \dfrac{4}{5}$

⑤ $\coth(2\ln x) = \dfrac{e^{2\ln x} + e^{-2\ln x}}{e^{2\ln x} - e^{-2\ln x}} = \dfrac{x^2 + \frac{1}{x^2}}{x^2 - \frac{1}{x^2}} = \dfrac{x^4 + 1}{x^4 - 1}$

Ex: Evaluate the derivative

① $y = \cosh x^4 \rightarrow \dfrac{dy}{dx} = 4x^3 \sinh x^4$

② $y = \ln(\tanh 2x) \rightarrow \dfrac{dy}{dx} = \dfrac{2\operatorname{sech}^2 2x}{\tanh 2x}$

③ $y = \sinh(\cos 3x) \rightarrow \dfrac{dy}{dx} = \cosh(\cos 3x) \cdot -3\sin 3x$

④ $y = \coth(\ln x) \rightarrow \dfrac{dy}{dx} = -\operatorname{csch}^2(\ln x) \cdot \dfrac{1}{x}$

⑤ $y = \cos^{-1}(\cosh 3x) \rightarrow \dfrac{dy}{dx} = \dfrac{-1}{\sqrt{1 - \cosh^2 3x}} \cdot 3\sinh 3x$

Ex: Find

(1) $\int \frac{\cosh\sqrt{x}}{\sqrt{x}}\,dx$ $\quad z=\sqrt{x}$, $\frac{dz}{2}=\frac{2\sqrt{x}}{2}dx$

$= 2\int \cosh z\,dz = 2\sinh z + C = 2\sinh\sqrt{x} + C$

(2) $\int \frac{\sinh 2x}{1+\cosh^2 2x}\,dx \underset{z=\cosh 2x}{=} \frac{1}{2}\int \frac{dz}{1+z^2} = \frac{1}{2}\tan^{-1}(\cosh 2x)+C$

(3) $\int \coth x\,\operatorname{csch}^3 x\,dx = \int \operatorname{csch}^2 x \cdot \operatorname{csch} x \coth x\,dx$ $\quad z=\operatorname{csch} x$

$= -\int z^2\,dz = -\frac{z^3}{3}+C = -\frac{\operatorname{csch}^3 x}{3}+C.$

(4) $\int \tanh 5x\,dx = \int \frac{\sinh 5x}{\cosh 5x}\,dx = \frac{1}{5}\ln|\cosh 5x| + C$

(5) $\int \frac{\pi^{\tanh 3x}}{\cosh^2 3x}\,dx = \int \pi^{\tanh 3x}\cdot \operatorname{sech}^2 3x\,dx$ $\quad z=\tanh 3x$

$= \int \pi^z \cdot \frac{1}{3}\,dz = \frac{1}{3}\frac{\pi^z}{\ln\pi}+C = \frac{1}{3\ln\pi}\pi^{\tanh 3x}+C.$

Inverse hyperbolic functions

$y = \sinh x \qquad D:(-\infty,\infty) \qquad R:(-\infty,\infty)$

$x = \sinh y = \dfrac{e^{y} - e^{-y}}{2} \iff e^{2y} - 2xe^{y} - 1 = 0$

$e^{y} = \dfrac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$, $e^{y} > 0$ and $\sqrt{x^2+1} > x$ $\rightarrow x - \sqrt{x^2+1} < 0$

$\Rightarrow e^{y} = x + \sqrt{x^2+1} \rightarrow y = \ln\left(x + \sqrt{x^2+1}\right)$

$\rightarrow \sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right) \qquad D:(-\infty,\infty) = \mathbb{R}$

inverse hyperbolic fcts

$\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right) \qquad D:(-\infty,\infty) \qquad R:(-\infty,\infty)$

$\cosh^{-1} x = \ln\left(x + \sqrt{x^2-1}\right) \qquad D:[1,\infty) \qquad R:(-\infty,\infty)$

$\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) \qquad D:(-1,1) \qquad R:(-\infty,\infty)$

$\coth^{-1} x = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right) \qquad D:(-\infty,-1)\cup(1,\infty) \qquad R:(-\infty,\infty)\setminus\{0\}$

$= \tanh\left(\frac{1}{x}\right)$

$\operatorname{sech}^{-1} x = \cosh^{-1}\left(\frac{1}{x}\right) = \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right) \qquad D:(0,1] \qquad R:(-\infty,\infty)$

$\operatorname{csch}^{-1} x = \sinh^{-1}\left(\frac{1}{x}\right) = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right) \qquad D:(-\infty,\infty)\setminus\{0\} = \mathbb{R}$

Examples

Find 1) $\sinh^{-1} 1$ 2) $\tanh^{-1}\left(\frac{1}{2}\right)$ 3) $\operatorname{csch}^{-1}(-\sqrt{3})$

Derivatives :

f	f'	
$\sinh^{-1} x$	$\frac{1}{\sqrt{x^2+1}}$	
$\cosh^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$	$, x>1$
$\tanh^{-1} x$	$\frac{1}{1-x^2}$	$, \lvert x\rvert<1$
$\coth^{-1} x$	$\frac{1}{1-x^2}$	$, \lvert x\rvert>1$
$\operatorname{sech}^{-1} x$	$\frac{-1}{x\sqrt{1-x^2}}$	$, 0<x<1$
$\operatorname{csch}^{-1} x$	$\frac{-1}{\lvert x\rvert\sqrt{1+x^2}}$	$, x\neq 0$

Integration

1) $\int \frac{dx}{\sqrt{a^2+x^2}} = \sinh^{-1}\left(\frac{x}{a}\right)+c$
$= \ln\left(x+\sqrt{x^2+a^2}\right)+c$

2) $\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\left(\frac{x}{a}\right)+c$
$= \ln\left(x+\sqrt{x^2-a^2}\right)+$

3) $\int \frac{dx}{a^2-x^2} = \frac{1}{a}\tanh^{-1}\frac{x}{a}+c$, if $\lvert x\rvert<a$
$= \frac{1}{a}\coth^{-1}\frac{x}{a}+c$, if $\lvert x\rvert>a$
$= \frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert+c$

4) $\int \frac{dx}{x\sqrt{a^2-x^2}} = -\frac{1}{a}\operatorname{sech}^{-1}\frac{x}{a}+c$
$= -\frac{1}{a}\ln\left(\frac{a+\sqrt{a^2-x^2}}{\lvert x\rvert}\right)$

5) $\int \frac{dx}{x\sqrt{a^2+x^2}} = \frac{-1}{a}\operatorname{csch}^{-1}\left\lvert\frac{x}{a}\right\rvert+c$
$= -\frac{1}{a}\ln\left(\frac{a+\sqrt{a^2+x^2}}{\lvert x\rvert}\right)$

Ex: Find $\frac{dy}{dx}$

1) $y=\cosh^{-1}\left(\frac{2x}{3}\right) \rightarrow \frac{dy}{dx} = \frac{2}{\sqrt{4x^2-9}}$

2) $y=\sinh^{-1}(\tanh x) \rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{\tanh^2 x+1}}\cdot \operatorname{sech}^2 x$

3) $y=\operatorname{sech}^{-1}\sqrt{x} \rightarrow \frac{dy}{dx} = \frac{-1}{\sqrt{x}\sqrt{1-x}}\cdot\frac{1}{2\sqrt{x}} = \frac{-1}{2x\sqrt{1-x}}$

4) $y=\tanh^{-1}(\coth^{-1}x) \rightarrow \frac{dy}{dx} = \frac{1}{1-(\coth^{-1}x)^2}\cdot\frac{1}{1-x^2}$

Ex: Find

1) $\int \frac{dx}{\sqrt{4x^2-9}} = \frac{1}{2}\int \frac{dz}{\sqrt{z^2-9}} = \frac{1}{2}\cosh^{-1}\left(\frac{z}{3}\right)+c = \frac{1}{2}\cosh^{-1}\frac{2x}{3}+c$
$= \frac{1}{2}\ln\left(2x+\sqrt{4x^2-9}\right)+c'$

2) $\int \frac{dx}{x\sqrt{9+4x^2}} = \frac{1}{3}\operatorname{sech}^{-1}\left(\frac{2x}{3}\right)+c$

3) $\int \frac{\sin x}{\sqrt{1+\cos^2 x}}\,dx = -\sinh^{-1}(\cos x)+c$

Integration by Parts

$$(f(x)g(x))' = f'g + g'f$$

integrate both sides $\rightarrow$ $fg = \int f'g + \int g'f$.

$\rightarrow$ $\int fg' = fg - \int f'g$

let $u = f(x)$, $dv = g'(x)\,dx$

$du = f'(x)dx$, $v = g(x)$

we have

$$\int u\,dv = uv - \int v\,du$$

Examples:

① $\int \frac{\ln x}{\sqrt{x}}\,dx$ $\quad u = \ln x \rightarrow du = \frac{1}{x}dx$ $\quad dv = x^{-\frac{1}{2}}dx \rightarrow v = 2\sqrt{x}$ (integrate -5)

$= 2\sqrt{x}\ln x - \int \frac{2\sqrt{x}}{x}\,dx = 2\sqrt{x}\ln x - 2\int x^{-\frac{1}{2}}dx$

$= 2\sqrt{x}\ln x - 4\sqrt{x} + C$.

② $\int_0^1 \tan^{-1}x\,dx$ $\quad u = \tan^{-1}x$ $\quad dv = (1)\,dx$

$du = \frac{1}{1+x^2}dx$ $\quad v = x$

$= x\tan^{-1}x - \int \frac{x}{1+x^2}\,dx$

$= x\tan^{-1}x - \frac{1}{2}\ln|1+x^2|\Big|_0^1 = \left(1\tan^{-1}1 - \frac{1}{2}\ln 2\right) - \left(0 - \frac{1}{2}\ln 1\right)$

$= \frac{\pi}{4} - \ln\sqrt{2}$.

③ $\int x\tan^2 2x\,dx$ $\quad u = x \quad dv = \tan^2 2x = (\sec^2 2x - 1)dx$

$du = dx \quad v = \frac{\tan 2x}{2} - x$ (arrow $-\int$)

$= x\left(\frac{\tan 2x}{2} - x\right) - \int \frac{\tan 2x}{2} - x\,dx$

$= \frac{1}{2}x\tan 2x - x^2 - \frac{1}{4}\ln|\sec 2x| + \frac{x^2}{2} + C$

$= \frac{1}{2}x\tan 2x - \frac{1}{2}x^2 - \frac{1}{4}\ln|\sec 2x| + C$

④ $\int \ln(x^2+4)\,dx$ $\quad u = \ln(x^2+4) \quad dv = dx$

$du = \frac{2x}{x^2+4}dx, \quad v = x$

$= x\ln(x^2+4) - \int \frac{2x^2}{x^2+4}dx$

$\begin{array}{r} 2 \\ x^2+4 \overline{)\,2x^2} \\ -2x^2+8 \\ \hline -8 \end{array}$

$= x\ln(x^2+4) - \int 2 - \frac{8}{x^2+4}\,dx$

$= x\ln(x^2+4) - 2x + \frac{8}{2}\tan^{-1}\left(\frac{x}{2}\right) + C$

$= x\ln(x^2+4) - 2x + 4\tan^{-1}\left(\frac{x}{2}\right) + C$

⑤ $\int_2^4 \sec^{-1}\sqrt{x}\,dx$ $\quad u = \sec^{-1}\sqrt{x} \quad dv = dx$

$du = \frac{1}{\sqrt{x}\sqrt{x-1}}\cdot\frac{1}{2\sqrt{x}} \quad v = x$

$= x\sec^{-1}\sqrt{x} - \frac{1}{2}\int \frac{1}{\sqrt{x-1}}dx$

$= x\sec^{-1}\sqrt{x} - \frac{1}{2}\cdot 2\sqrt{x-1}\Big|_2^4 = \left(4\frac{\pi}{3} - \sqrt{3}\right) - \left(2\frac{\pi}{4} - 1\right) = \frac{5\pi}{6} - \sqrt{3} + 1$

⑥ $\int \sin(\ln x)\,dx$

$u = \sin(\ln x) \quad dv = dx$

$du = \cos(\ln x)\cdot\frac{1}{x} \quad v = x$

$= x\sin(\ln x) - \int \cos(\ln x)\,dx$

$u = \cos(\ln x) \quad dv = dx$

$du = -\sin(\ln x)\frac{1}{x} \quad v = x$

$= x\sin(\ln x) - \left[x\cos(\ln x) + \int \sin(\ln x)dx\right]$

$= x\sin(\ln x) - x\cos(\ln x) - \int \sin(\ln x)\,dx$

$\Rightarrow \int \sin(\ln x)\,dx = \frac{1}{2}\left[x\sin(\ln x) - x\cos(\ln x)\right]$

Integration by Parts
Tabular method

$$\int P_n(x)\, f(x)\, dx$$

$P(x)$ is polynomial of order n

$f(x)$ f_n can be integrated $(n+1)$ times

$\oplus$ $P_n(x)$ — $f(x)$

$\ominus$ $P_n'(x)$ — $\int f = f_1$

$\oplus$ P_n'' — $\int f_1 = f_2$

$\vdots$

$P_n^{(n+1)} = 0$ — $\int f_n = f_{n+1}$

$$\int P_n(x) f(x)\, dx = +P_n f_1 - P_n' f_2 + P_n'' f_3 - \cdots \pm P_n^{(n)} f_{n+1} + C$$

Examples:

① $\int x^3 \sin 2x\, dx$

$\oplus$ x^3 — $\sin 2x$

$\ominus$ $3x^2$ → $-\frac{\cos 2x}{2}$

$\oplus$ $6x$ → $-\frac{\sin 2x}{4}$

$\ominus$ 6 → $\frac{\cos 2x}{8}$

0 → $\frac{\sin 2x}{16}$

$$= -\frac{1}{2}x^3\cos 2x + \frac{3}{4}x^2 \sin 2x + \frac{6}{8}x\cos 2x - \frac{6}{16}\sin 2x$$

$$= \left(\frac{3}{4}x^2 - \frac{3}{8}\right)\sin 2x + \left(\frac{3}{4}x - \frac{1}{2}x^3\right)\cos 2x + C$$

(2) $\int \frac{3x^2-x+2}{e^{2x}}\,dx = \int (3x^2-x+2)e^{-2x}\,dx$

+	$3x^2-x+2$	e^{-2x}
−	$6x-1$	$\frac{e^{-2x}}{-2}$
+	6	$\frac{e^{-2x}}{4}$
	0	$\frac{e^{-2x}}{-8}$

$= -\frac{1}{2}(3x^2-x+2)e^{-2x} - \frac{1}{4}(6x-1)e^{-2x} - \frac{6}{8}e^{-2x} + c$

(3) $\int x^3 \cos x^2\,dx$

$z = x^2$

$dz = 2x\,dx$

$\therefore = \int x \cdot x^2 \cos x^2\,dx$

$= \int z \cos z \frac{1}{2}\,dz$

+	z	$\cos z$
−	1	$\sin z$
	0	$-\cos z$

$= \frac{1}{2}\left[x^2 \sin x^2 + \cos x^2\right] + c$

(4) $\int x^2\sqrt{x-1}\,dx$

+	x^2	$(x-1)^{1/2}$
−	$2x$	$\frac{2}{3}(x-1)^{3/2}$
+	2	$\frac{2}{35}(x-1)^{5/2}$
	0	$\frac{4}{15}\frac{2}{7}(x-1)^{7/2}$

$= \frac{2}{3}x^2(x-1)^{3/2} - \frac{8}{15}x(x-1)^{5/2} + \frac{16}{105}(x-1)^{7/2} + c$

Trigonometric integrals

* Remember

$\sin^2\theta + \cos^2\theta = 1$ → $\sin^2\theta = 1 - \cos^2\theta$, $\cos^2\theta = 1 - \sin^2\theta$

and $\sin^2\theta = \frac{1-\cos 2\theta}{2}$, $\cos^2\theta = \frac{1+\cos 2\theta}{2}$

* $\int \sin^n x\, dx$ — n odd: $\sin^2 x = 1 - \cos^2 x$ and let $z = \cos x$; n even: $\sin^2 x = \frac{1-\cos 2x}{2}$

or use $\int \sin^n x\, dx = \frac{-1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x\, dx$

* $\int \cos^n x\, dx$ — n odd: $\cos^2 x = 1 - \sin^2 x$ and let $z = \sin x$; n even: $\cos^2 x = \frac{1+\cos 2x}{2}$.

or use $\int \cos^n x\, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x\, dx$

Examples

(1) $\int \sin^2 3x\,dx = \int \frac{1+\cos 6x}{2}\,dx = \frac{1}{2}\left[x + \frac{\sin 6x}{6}\right] + C$

(2) $\int \cos^3 2x\,dx = \int \cos^2 2x \cos 2x\,dx = \int (1-\sin^2 2x)\cos 2x\,dx$

$z = \sin 2x,\ dz = 2\cos 2x \rightarrow \int (1-z^2)\frac{1}{2}\,dz = \frac{1}{2}\left(z - \frac{z^3}{3}\right)$

$= \frac{1}{2}\sin 2x - \frac{1}{6}\sin^3 2x + C$

or by law $\int \cos^3 2x\,dx \underset{z=2x}{=\!=} \frac{1}{2}\int \cos^3 z\,dz = \frac{1}{2}\left[\frac{1}{3}\cos^2 z \sin z + \frac{2}{3}\int \cos z\,dz\right]$

$= \frac{1}{6}\cos^2 z \sin z + \frac{1}{3}\sin z + C = \frac{1}{6}\cos^2 2x \sin 2x + \frac{1}{3}\sin 2x + C$

$= \frac{1}{6}(1-\sin^2 2x)\sin 2x + \frac{1}{3}\sin 2x + C = \frac{1}{2}\sin 2x - \frac{1}{6}\sin^3 2x + C$

(3) $\int \sin^4 5x\,dx = \int \sin^2 5x \sin^2 5x\,dx = \int \left(\frac{1-\cos 10x}{2}\right)\left(\frac{1-\cos 10x}{2}\right)$

$= \frac{1}{4}\int 1 - 2\cos 10x + \cos^2 10x\,dx = \frac{1}{2}\int 1 - 2\cos 10x + \frac{1+\cos 20x}{2}\,dx$

$= \frac{1}{2}\left[x - \frac{2\sin 10x}{10} + \frac{1}{2}x + \frac{\sin 20x}{40}\right] + C$

$= \frac{1}{2}\left[\frac{3}{2}x - \frac{1}{5}\sin 10x + \frac{1}{40}\sin 20x\right] + C$

$$* \int \sin^n x \cos^m x \, dx$$

if n odd use $\sin^2 x = 1 - \cos^2 x$, $z = \cos x$

or if m odd " $\cos^2 x = 1 - \sin^2 x$, $z = \sin x$

if n and m are even use

$\sin^2 x = \frac{1 - \cos 2x}{2}$ and $\cos^2 x = \frac{1 + \cos 2x}{2}$.

Examples:

(1) $\int \sin^4 3x \cos^3 3x \, dx = \int \sin^4 3x \cos^2 3x \cos 3x \, dx$

$= \int \sin^4 3x (1 - \sin^2 3x) \cos 3x \, dx$, $z = \sin 3x$, $dz = \frac{1}{3}\cos 3x \, dx$

$= \int z^4 (1 - z^2) \frac{1}{3} dz = \frac{1}{3} \int z^4 - z^6 \, dz = \frac{1}{3}\left(\frac{z^5}{5} - \frac{z^7}{7}\right) + C$

$= \frac{\sin^5 3x}{15} - \frac{\sin^7 3x}{21} + C$

(2) $\int \sin^3 4x \cos^3 4x \, dx = \int \sin^2 4x \cos^3 4x \sin 4x \, dx = \int (1 - \cos^2 4x) \cos^3 4x \sin 4x \, dx$

$\underset{z = \cos 4x}{=} \int (1 - z^2) z^3 \frac{1}{4} dz = \frac{1}{4} \int z^3 - z^5 dz = \frac{1}{4}\left(\frac{\cos^4 4x}{4} - \frac{\cos^6 4x}{6}\right) + C$

(3) $\int \sin^2 2x \cos^2 2x \, dx = \int (\sin 2x \cos 2x)^2 dx = \int \left(\frac{1}{2}\sin 4x\right)^2$

$= \frac{1}{4} \int \sin^2 4x \, dx = \frac{1}{4} \int \frac{1 - \cos 8x}{2} dx = \frac{1}{8}\left(x - \frac{\sin 8x}{8}\right) + C$

* $\int \sin mx \cos nx \, dx$, $\int \sin mx \sin nx \, dx$, $\int \cos mx \cos nx \, dx$

Remember

$\sin\alpha\cos\beta = \frac{1}{2}\left[\sin(\alpha-\beta) + \sin(\alpha+\beta)\right]$

$\sin\alpha\sin\beta = \frac{1}{2}\left[\cos(\alpha-\beta) - \cos(\alpha+\beta)\right]$

$\cos\alpha\cos\beta = \frac{1}{2}\left[\cos(\alpha-\beta) + \cos(\alpha+\beta)\right]$

Examples :

(1) $\int \sin 7x \cos 3x \, dx = \frac{1}{2}\int \sin 4x + \sin 10x \, dx = \frac{1}{2}\left[\frac{-\cos 4x}{4} - \frac{\cos 10x}{10}\right]$

(2) $\int \cos 3x \sin 2x \, dx \overset{\substack{\alpha=2 \\ \beta=3}}{=\!=} \frac{1}{2}\int \sin(-x) + \sin 5x \, dx = \frac{1}{2}\int -\sin x + \sin 5x \, dx$

$= \frac{1}{2}\left[\cos x - \frac{\cos 5x}{5}\right] + C$.

(3) $\int \cos 2x \cos 4x \, dx = \frac{1}{2}\int \cos 2x + \cos 6x \, dx = \frac{1}{2}\left[\frac{\sin 2x}{2} + \frac{\sin 6x}{6}\right] + C$

remember $\cos(-\theta) = \cos\theta$

(4) $\int \sin^2 2x \cos^2 3x \, dx = \int \left(\sin 2x \cos 3x\right)^2 dx$

$= \int \left(\frac{\sin(-x) + \sin 5x}{2}\right)^2 = \frac{1}{4}\int \sin^2 x + 2\sin x \sin 5x + \sin^2 5x \, dx$

$= \frac{1}{4}\int \frac{1-\cos 2x}{2} - \left(\cos 4x - \cos 6x\right) + \frac{1-\cos 10x}{2} \, dx$

$= \frac{1}{4}\left[x - \frac{\sin 2x}{4} - \frac{\sin 4x}{4} + \frac{\sin 6x}{6} - \frac{\sin 10x}{20}\right] + C$.

* $\int \tan^n x\,dx$, $\int \sec^n x\,dx$

$$\int \tan^n x\,dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x\,dx .$$

$$\int \sec^n x\,dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx .$$

Examples:

(1) $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\ln|\cos x| + c = \ln|\sec x| + c$

(2) $\int \sec x\,dx \cdot \frac{\sec x + \tan x}{\sec x + \tan x} = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\,dx = \ln|\sec x + \tan x| + c$

(3) $\int \tan^2 x\,dx = \int \sec^2 x - 1\,dx = \tan x - x + c$.

(4) $\int \sec^2 x\,dx = \tan x + c$

(5) $\int \tan^3 x\,dx = \int \tan x(\sec^2 x - 1)\,dx = \int \underbrace{\tan x}_{z} \underbrace{\sec^2 x\,dx}_{dz} - \int \tan x$

$= \frac{\tan^2 x}{2} - \ln|\sec x| + c$

(6) $\int \sec^3 x\,dx = \int \sec x \sec^2 x\,dx$ $\quad u = \sec x \quad dv = \sec^2 x$ $\quad du = \sec x \tan x \quad v = \tan x$

$\therefore \int \sec^3 x\,dx = \sec x \tan x - \int \sec x \tan^2 x\,dx = \sec x \tan x - \int \sec x(\sec^2 x - 1)\,dx$

$\therefore \int \sec^3 x = \sec x \tan x - \int \sec^3 x\,dx + \int \sec x\,dx$

$2\int \sec^3 x\,dx = \sec x \tan x + \ln|\sec x + \tan x| + c$

$\therefore \int \sec^3 x\,dx = \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + c$ OR we can use the formula

$$* \int \tan^m x \sec^n x \, dx$$

If m odd use $\tan^2 x = \sec^2 x - 1$ and let $z = \sec x$

If n even use $\sec^2 x = \tan^2 x - 1$ and $z = \tan x$

If m even and n odd use $\tan^2 x = \sec^2 x - 1$

and use $\int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$

Examples:

(1) $\int \sec^4 x \tan^2 x \, dx = \int \tan^2 x (1 + \tan^2 x) \sec^2 x \, dx$ $\boxed{z = \tan x}$

$= \int z^2 (1 + z^2) \, dz = \int z^2 + z^4 \, dz = \frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C$

(2) $\int \tan^3 x \sec x \, dx = \int \tan^2 x \sec^2 x \cdot \sec x \tan x \, dx$ $\boxed{z = \sec x}$

$= \int (\sec^2 x - 1) \sec^2 x \cdot \sec x \tan x \, dx = \int (z^2 - 1) z^2 \, dz$

$= \frac{z^5}{5} - \frac{z^3}{3} + C = \frac{\sec^5 x}{5} - \frac{\sec^3 x}{3} + C$

(3) $\int \tan^2 x \sec x \, dx = \int (\sec^2 x - 1) \sec x \, dx$

$= \int \sec^3 x \, dx - \int \sec x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln|\sec x + \tan x| - \ln|\sec x + \tan x|$

$= \frac{1}{2} \sec x \tan x - \frac{1}{2} \ln|\sec x + \tan x| + C$.

Trigonometric Substitutions

Any integral contain the following expressions we use the corresponding substitution

$a^2 - z^2$	$a^2 + z^2$	$z^2 - a^2$
# z (f= in x)		
$z = a\sin\theta$	$z = a\tan\theta$	$z = a\sec\theta$
$dz = a\cos\theta\, d\theta$	$dz = a\sec^2\theta\, d\theta$	$dz = a\sec\theta\tan\theta\, d\theta$
and use	and use	and use
$1 - \sin^2\theta = \cos^2\theta$	$1 + \tan^2\theta = \sec^2\theta$	$\sec^2\theta - 1 = \tan^2\theta$

z here Function in x

Ex: Write only Trigonometric substitution

① $9 - 4x^2 \longrightarrow 9 - (2x)^2 \xrightarrow[a=3]{z=2x}$ $2x = 3\sin\theta$, $2\,dx = 3\cos\theta\, d\theta$

② $e^{2x} + 1 \longrightarrow (e^x)^2 + 1 \xrightarrow[a=1]{z=e^x}$ $e^x = \tan\theta$, $e^x\, dx = \sec^2\theta\, d\theta$

③ $3x^2 - 1 \longrightarrow$ $\sqrt{3}\,x = \sec\theta$, $\sqrt{3}\,dx = \sec\theta\tan\theta\, d\theta$

the integral will be in term of θ

integrate then replace for x.

Examples

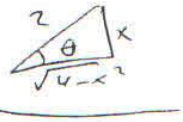
(1) $\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}}$, $x = 2\sin\theta$, $dx = 2\cos\theta\, d\theta$

$= \int \frac{2\cos\theta\, d\theta}{4\sin^2\theta\sqrt{4-4\sin^2\theta}} = \int \frac{2\cos\theta}{4\sin^2\theta \cdot 2\cos\theta}\, d\theta$

$= \frac{1}{4}\int \csc^2\theta\, d\theta = \frac{-1}{4}\cot\theta$

$= \frac{-1}{4}\frac{\sqrt{4-x^2}}{x}\Big|_1^{\sqrt{2}} = \frac{-1}{4}\left[\left(\frac{\sqrt{2}}{\sqrt{2}}\right) - \left(\sqrt{3}\right)\right] = \frac{\sqrt{3}-1}{4}$

$x = 2\sin\theta$
$\sin\theta = \frac{x}{2}$



(2) $\int \frac{\sqrt{x^2-25}}{x}\, dx$, $x = 5\sec\theta$, $dx = 5\sec\theta\tan\theta\, d\theta$

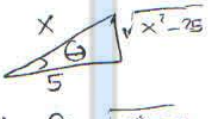
$= \int \frac{\sqrt{25\sec^2\theta - 25}}{5\sec\theta} \cdot 5\sec\theta\tan\theta\, d\theta = \int 5\tan^2\theta\, d\theta$

$= 5\int \sec^2\theta - 1\, d\theta = 5(\tan\theta - \theta) + C$

$= 5\left(\frac{\sqrt{x^2-25}}{5} - \sec^{-1}\left(\frac{x}{5}\right)\right) + C$

$= \sqrt{x^2-25} - 5\sec^{-1}\left(\frac{x}{5}\right) + C$

$\sec\theta = \frac{x}{5}$



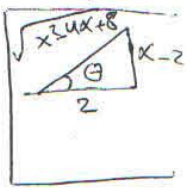
$\tan\theta = \frac{\sqrt{x^2-25}}{5}$

(3) $\int \frac{x}{x^2-4x+8}\, dx = \int \frac{x}{x^2-4x+4+4}\, dx = \int \frac{x}{(x-2)^2+4}\, dx$

$= \int \frac{2\tan\theta + 2}{4\tan^2\theta + 4} \cdot 2\sec^2\theta\, d\theta = \int \tan\theta + 1\, d\theta = \ln|\sec\theta| + \theta + C$

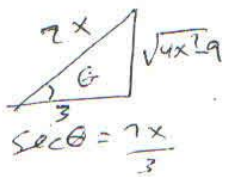
$= \ln\left|\frac{\sqrt{x^2-4x+8}}{2}\right| + \tan^{-1}\left(\frac{x-2}{2}\right) + C$

$x - 2 = 2\tan\theta$
$dx = 2\sec^2\theta\, d\theta$



(4) $\int \frac{dx}{(4x^2-9)^{3/2}}$ $\quad 2x = 3\sec\theta$, $2dx = 3\sec\theta\tan\theta\, d\theta$

$= \int \frac{\frac{3}{2}\sec\theta\tan\theta\, d\theta}{(9\sec^2\theta - 9)^{3/2}} = \frac{3}{2}\int \frac{\sec\theta\tan\theta}{3^3\tan^3\theta} = \frac{1}{18}\int \frac{\cos\theta}{\sin^2\theta}\, d\theta$



$\sec\theta = \frac{2x}{3}$

$= \frac{1}{18}\int \csc\theta\cot\theta = \frac{-1}{18}\csc\theta + C$

$= \frac{-1}{18}\cdot\frac{2x}{\sqrt{4x^2-9}} + C$

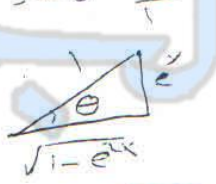
(5) $\int e^x\sqrt{1-e^{2x}}\, dx$

$e^x = \sin\theta \rightarrow e^x dx = \cos\theta\, d\theta$

$= \int \sqrt{1-\sin^2\theta}\cdot\cos\theta\, d\theta = \int \cos^2\theta\, d\theta = \int \frac{1+\cos 2\theta}{2}\, d\theta$

$= \frac{1}{2}\left[\theta + \frac{\sin 2\theta}{2}\right] = \frac{1}{2}\left[\theta + \frac{2\sin\theta\cos\theta}{2}\right] + C$

$= \frac{1}{2}\left[\sin^{-1}e^x + e^x\sqrt{1-e^{2x}}\right] + C$



or $\cos\theta = \sqrt{1-\sin^2\theta}$

(6) $\int \sqrt{x(4-x)}\, dx$

$= \int \sqrt{4x-x^2}\, dx = \int \sqrt{-(x^2-4x+4-4)}\, dx = \int \sqrt{4-(x-2)^2}\, dx$

i.e. $\sqrt{4-x^2}$

$x-2 = 2\sin\theta \rightarrow dx = 2\cos\theta\, d\theta \rightarrow \int \frac{\sqrt{4-4\sin^2\theta}}{2\cos\theta}\cdot 2\cos\theta\, d\theta$

$= \int 4\cos^2\theta\, d\theta = 4\int \frac{1+\cos 2\theta}{2}\, d\theta$

$= 2\left(\theta + \frac{\sin 2\theta}{2}\right) = 2(\theta + \sin\theta\cos\theta) = 2\left(\sin^{-1}\left(\frac{x-2}{2}\right) + \frac{(x-2)}{2}\cdot\frac{\sqrt{4x-x^2}}{2}\right)$

$= 2\sin^{-1}\left(\frac{x-2}{2}\right) + \frac{(x-2)}{2}\sqrt{4x-x^2} + C$

Partial fractions

$\int \frac{P(x)}{Q(x)}\,dx$ where $P(x)$ and $Q(x)$ are polynomials

degree $P(x)$ < degree $Q(x)$

(if deg. $P(x) \geq$ deg. $Q(x)$ use long division)

we should make partial fraction on $\frac{P(x)}{Q(x)}$ which depend on $Q(x)$.

Case 1: Distinct linear factors

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(a_1x+b_1)(a_2x+b_2)\cdots(a_nx+b_n)} = \frac{A_1}{a_1x+b_1} + \frac{A_2}{a_2x+b_2} + \cdots + \frac{A_n}{a_nx+b_n}$$

Ex: $\frac{2x-1}{x^3-4x^2} = \frac{2x-1}{x(x-2)(x+2)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+2}$.

Case 2: repeated linear factors

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(ax+b)^n} = \frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_n}{(ax+b)^n}$$

Ex: $\frac{3x^2}{x^3-x^2-x+1} = \frac{3x^2}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$.

Case 3: irreducible quadratic factor

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{ax^2+bx+c} = \frac{A_1x+B_1}{ax^2+bx+c}$$

$\hookrightarrow \Delta = b^2-4ac < 0$

Ex: $\frac{x-1}{\cdots} = \frac{Ax+B}{\cdots}$

Case 4: repeated irreducible factors.

$$\frac{P(x)}{q(x)} = \frac{P(x)}{(ax^2+bx+c)^n} = \frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \cdots + \frac{A_nx+B_n}{(ax^2+bx+c)^n}$$

Ex: $$\frac{2x^2+1}{x^2(x+2)(x^2+3)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2} + \frac{Dx+E}{x^2+3} + \frac{Fx+G}{(x^2+3)^2}.$$

Ex: Write out the form of partial fraction decomposition.

a) $$\int \frac{x^3+5}{(x^2-x-2)(x^4-16)}\,dx$$

$$\frac{x^3+5}{(x+1)(x-2)(x^2+4)(x^2+4)} = \frac{x^3+5}{(x+1)(x+2)(x-2)^2(x^2+4)} = \frac{A_1}{x+1} + \frac{A_2}{x+2} + \frac{A_3}{(x-2)} + \frac{A_4}{(x-2)^2} + \frac{A_5x+A_6}{x^2+4}.$$

b) $$\int \frac{2x^2-1}{(x^3+2x)(2x^2+x)}\,dx$$

$$\frac{2x^2-1}{x(x^2+2)\cdot x(2x+1)} = \frac{2x^2-1}{x^2(x^2+2)(2x+1)}$$

$$= \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+2} + \frac{E}{2x+2}.$$

Ex 1: Evaluate $\int \frac{5x-10}{x^2-3x-4}\,dx$

$$\frac{5x-10}{x^2-3x-4} = \frac{5x-10}{(x-4)(x+1)} = \frac{A}{x-4} + \frac{B}{x+1}$$

$$5x-10 = A(x+1) + B(x-4)$$

$x=4 \rightarrow 10 = 5A \rightarrow \boxed{A=2}$

$x=-1 \rightarrow -15 = -5B \rightarrow \boxed{B=3}$

$$\therefore \int \frac{5x-10}{x^2-3x-4}\,dx = \int \frac{2}{x-4} + \frac{3}{x+1}\,dx = 2\ln|x-4| + 3\ln|x+1| + C$$

Ex 2: $\int \frac{2x+4}{x^3-2x^2}\,dx$

$$\frac{2x+4}{x^2(x-2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2}$$

$$2x+4 = Ax(x-2) + B(x-2) + Cx^2$$

$x=0 \rightarrow +4 = -2B \rightarrow \boxed{B=-2}$

$x=2 \rightarrow 8 = 4C \rightarrow \boxed{C=2}$

$x=1 \rightarrow 6 = -A-B+C \rightarrow \boxed{A=-2}$

$$\Rightarrow \int \frac{2x+4}{x^3-2x^2}\,dx = \int \frac{-2}{x} - \frac{2}{x^2} + \frac{2}{x-2}\,dx = -2\ln|x| + \frac{2}{x} + 2\ln|x-2|$$
$$= \ln\left(\frac{x-2}{x}\right)^2 + \frac{2}{x} + C$$

Ex 3: $\int \frac{2(x^2-x+1)}{x^3+2x}\,dx$

$$\frac{2(x^2-x+1)}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2} \Rightarrow 2(x^2-x+1) = A(x^2+2) + (Bx+C)x$$

$x=0 \rightarrow A=1$

$x=1 \rightarrow 3A+B+C=2$

$x=-1 \rightarrow 3A+B-C=6$

$\boxed{A=1}$, $\boxed{B=1}$, $\boxed{C=-2}$

$$= \int \frac{1}{x} + \frac{x-2}{x^2+2}\,dx$$

$$= \int \frac{1}{x} + \frac{x}{x^2+2} - \frac{2}{x^2+2}\,dx = \ln|x| + \frac{1}{2}\ln|x^2+2| - \frac{2}{\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C$$

Ex4: $\int \sqrt{1+e^x}\,dx$

$z=\sqrt{1+e^x}$, $dz=\frac{e^x}{2\sqrt{1+e^x}}dx \rightarrow \frac{2z}{z^2-1}dz=dx$

$\int \sqrt{1+e^x}\,dx = \int z\cdot\frac{2z}{z^2-1}dz = \int \frac{2z^2}{z^2-1}dz$

$= \int 2+\frac{2}{z^2-1}dz = \int 2+\frac{1}{z-1}-\frac{1}{z+1}\,dz$

$= 2z+\ln|z-1|-\ln|z+1|+C = 2z+\ln\left|\frac{z-1}{z+1}\right|+C$

$= 2\sqrt{1+e^x}+\ln\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right|+C$

Ex 5: $\int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2}\,dx$

use long division

$= \int 3x^2+1+\frac{1}{x^2+x-2}\,dx = \int 3x^2+1+\frac{1/3}{x-1}+\frac{-1/3}{x+2}\,dx$

$= x^3+x+\frac{1}{3}\ln|x-1|-\frac{1}{3}\ln|x+2|+C$

$= x^3+x+\frac{1}{3}\ln\left|\frac{x-1}{x+2}\right|+C$

Ex: $\int \frac{dx}{x - x^{\frac{3}{5}}}$, $x = z^5$, $dx = 5z^4 dz$

$$= \int \frac{5z^4 dz}{z^5 - z^3} = \int \frac{5z^4}{z^3(z^2-1)} dz = 5\int \frac{5z}{z^2-1} dz$$

$$= \frac{5}{2} \ln|z^2-1| = \frac{5}{2} \ln|x^{\frac{2}{5}} - 1| + C.$$

Ex: $\int \frac{x}{\sqrt{x-1}} dx$

L'Hopital rule

Theorem:

Suppose f and g are differentiable functions on an open interval containing a, except possibly at a and $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$.

, then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$.

Examples:

Remember $e^{+\infty} \to +\infty$

(1) $\lim_{x\to\infty} \frac{x^2}{e^{3x}-1} = \frac{\infty}{\infty}$

$\therefore \lim_{x\to\infty} \frac{x^2}{e^{3x}-1} = \lim_{x\to\infty} \frac{2x}{3e^{3x}} = \lim_{x\to+\infty} \frac{2}{9e^{3x}} = 0$

(2) $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{0}{0}$

$\therefore \lim_{x\to 0} \frac{1-\cos x}{x^2} = \lim_{x\to 0} \frac{\sin x}{2x} = \lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2}$.

Remember $\ln 0^+ \to -\infty$

(3) $\lim_{x\to 0^+} \sin x \ln x = 0 \cdot -\infty$

$= \lim_{y\to 0^+} \frac{\ln x}{\frac{1}{\sin x}} = \lim_{x\to 0^+} \frac{\ln x}{\csc x} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\csc x \cot x} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{\cos x}{\sin^2 x}}$

$= -\lim_{x\to 0} \frac{\sin^2 x}{x\cos x} = -\lim_{x\to 0} \frac{2\sin x\cos x}{\cos x - x\sin x} = 0$

(4) $\lim_{x\to\frac{\pi}{4}} (1-\tan x)\sec 2x = 0\cdot\infty$

$= \lim_{x\to\frac{\pi}{4}} \frac{1-\tan x}{\cos 2x} = \lim_{x\to\frac{\pi}{4}} \frac{-\sec^2 x}{-2\sin 2x} = \frac{2}{2} = 1$

(5) $\lim_{x\to 0^+} \frac{1}{x} - \frac{1}{\sin x} = +\infty - \infty$

$= \lim_{x\to 0^+} \frac{\sin x - x}{x\sin x} = \lim_{x\to 0^+} \frac{\cos x - 1}{\sin x + x\cos x} = \lim_{x\to 0^+} \frac{-\sin x}{\cos x + \cos x - x\sin x} = 0$

(6) $\lim_{x\to+\infty} x - \ln(x^2+1) = \infty - \infty$

$= \lim_{x\to\infty} \ln e^x - \ln(x^2+1) = \lim_{x\to\infty} \ln\left(\frac{e^x}{x^2+1}\right)$

$= \ln\left(\lim_{x\to+\infty} \frac{e^x}{x^2+1}\right) = \ln\left(\lim \frac{e^x}{2x}\right) = \ln\left(\lim \frac{e^x}{2}\right) = \ln(+\infty) = +\infty$

Other indeterminate forms:

1^∞ , ∞^0 , 0^0

$L = \lim_{x\to a} f^g$

$\ln L = \ln \lim_{x\to a} f^g = \lim \ln f^g = \lim g \ln f = \lim_{x\to a} \frac{\ln f}{1/g} = L_1$

apply L'Hopital rule on $\ln L$

and then $L = e^{\ln L} = e^{L_1}$

Ex1: $\lim_{x\to+\infty} 3\left(x^{\frac{1}{x}}\right) = \infty^0$

$\lim \ln x^{\frac{1}{x}} = \lim_{x\to\infty} \frac{1}{x}\ln x = \lim \frac{\ln x}{x} = \lim_{x\to\infty} \frac{\frac{1}{x}}{1} = 0$

$\lim x^{\frac{1}{x}} = e^0 = 1 \rightarrow 3\lim x^{\frac{1}{x}} = 3(1) = 3$

Ex2: $\lim_{x\to 0} x^{x^2} = 0^0$

$\lim_{x\to 0} \ln x^{x^2} = \lim_{x\to 0} x^2 \ln x = \lim_{x\to 0} \frac{\ln x}{\frac{-2}{x^2}} = \lim_{x\to 0} \frac{\frac{1}{x}}{-2x^{-3}} = \lim_{x\to 0} \frac{\frac{1}{x}}{\frac{-2}{x^3}}$

$= \lim_{x\to 0} \frac{x^3}{-2x} = \lim_{x\to 0} \frac{x^2}{-2} = 0 \Rightarrow \lim_{x\to 0} x^{x^2} = e^0 = 1$

Ex3: $\lim_{x\to 0} (\sin x)^{\frac{3}{\ln x}}$

$\lim_{x\to 0} \ln (\sin x)^{\frac{3}{\ln x}} = \lim_{x\to 0} \frac{3}{\ln x} \ln(\sin x) = 3 \lim_{x\to 0} \frac{\ln \sin x}{\ln x}$

$= 3 \lim_{x\to 0} \frac{\frac{\cos x}{\sin x}}{\frac{1}{x}} = 3 \lim_{x\to 0} \frac{x \cos x}{\sin x} = 3 \Rightarrow \lim_{x\to 0} (\sin x)^{3/\ln x} = e^3$

Ex4: $\lim_{x\to +\infty} (\ln x)^{\frac{1}{x}}$

$\lim_{x\to \infty} \frac{\ln(\ln x)}{x} = \lim_{x\to \infty} \frac{\frac{1}{x}}{\ln x} = \lim_{x\to \infty} \frac{1}{x \ln x} = 0$

$\Rightarrow \lim_{x\to \infty} (\ln x)^{\frac{1}{x}} = e^0 = 1$

* $\lim_{x\to +\infty} \left(1 + \frac{a}{x}\right)^{bx} = \lim_{x\to 0} (1+ax)^{\frac{b}{x}} = e^{ab}$

Ex: $\lim_{x\to \infty} \left(1 + \frac{2}{x}\right)^{3x} = e^6$

Ex: $\lim_{x\to 0} (1+3x)^{\frac{1}{2x}} = e^{\frac{3}{2}}$

Ex: $\lim_{x\to +\infty} \left(1 - \frac{1}{2x}\right)^{\frac{x}{5}} = e^{\frac{-1}{10}} = \frac{1}{\sqrt[10]{e}}$

Improper integral

Type I: $\int_a^b f(x)\,dx$ is called improper integral

if $a = -\infty$ or $b = +\infty$ or both.

if the value of the integral $\int_a^b f(x)\,dx = L$ → integral converge

" " " " " $= \infty$ → integral diverge

we use limit to find the integral

Ex1: $\int_1^\infty \frac{dx}{x} = \lim_{b\to+\infty} \int_1^b \frac{dx}{x} = \lim_{b\to\infty} \ln|x| \Big|_1^b$

$= \lim_{b\to+\infty} \ln|b| - \ln 1 = \infty$ → integral diverge

Ex2: $\int_0^\infty (1-x)e^{-x}\,dx$

+	$1-x$	e^{-x}
−	-1	$-e^{-x}$
	0	e^{-x}

$= \lim_{b\to+\infty} -(1-x)e^{-x} + e^{-x} \Big|_0^b$

$= \lim_{b\to+\infty} x e^{-x} \Big|_0^b = \lim_{b\to+\infty} b e^{-b} = \lim_{b\to\infty} \frac{b}{e^b} = \lim_{b\to\infty} \frac{1}{e^b} = \frac{1}{\infty} = 0$

∴ integral converge to 0

Ex3: $\int_{-\infty}^0 \frac{x}{\sqrt{x^2+2}}\,dx = \lim \int \frac{x}{\sqrt{x^2+2}}\,dx = \lim \frac{1}{2}\int \frac{dz}{\sqrt{z}}$

$= \lim_{a\to+\infty} \sqrt{x^2+2} \Big|_a^0 = \sqrt{2} - \infty = -\infty$

→ integral diverge

Ex4: $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_{0}^{\infty} \frac{dx}{1+x^2}$

$= \lim_{a\to\infty} \int_{a}^{0} \frac{dx}{1+x^2} + \lim_{b\to+\infty} \int_{0}^{b} \frac{dx}{1+x^2}$

$= \lim_{a\to\infty} \tan^{-1}x \Big|_{a}^{0} + \lim \tan^{-1}x \Big|_{0}^{b} = \lim_{a\to+\infty} -\tan^{-1}a + \lim_{b\to+\infty} \tan^{-1}b$

$= -\left(\frac{-\pi}{2}\right) + \frac{\pi}{2} = \pi \Rightarrow$ integral converge to π

Type II

The integral $\int_a^b f(x)\,dx$ is said to be improper integral if $f(x)$ discontinuous at a or at b or at $c \in (a,b)$.

if f discont. at $a \Rightarrow \int_a^b f(x)\,dx = \lim_{k\to a^+} \int_k^b f(x)\,dx$

" " " $b \Rightarrow \int_a^b f(x)\,dx = \lim_{k\to b^-} \int_a^k f(x)\,dx$

" " " $c \in (a,b) \Rightarrow \int_a^b f(x)\,dx = \int_a^c f\,dx + \int_c^b f\,dx$

$= \lim_{k_1\to c^-} \int_a^{k_1} f\,dx + \lim_{k_2\to c^+} \int_{k_2}^b f\,dx$

Examples

Ex1: $\int_1^2 \frac{dx}{1-x} = \lim_{k\to 1^+} \int_k^2 \frac{dx}{1-x} = \lim_{k\to 1^+} -\ln|1-x| \Big|_k^2$

$= \lim_{k\to 1^+} -\ln 1 + \ln|1-k| = -\infty \Rightarrow$ integral diverge

Ex2: $\int_0^1 \frac{dx}{\sqrt{x}(x+1)}$ Let $z=\sqrt{x}$ $dz=\frac{1}{2\sqrt{x}}dx$

$2dz=\frac{dx}{\sqrt{x}}$

$= \lim_{k\to 0^+} \int_k^1 \frac{2dz}{z^2+1} = 2\lim_{k\to 0^+} \tan^{-1}\sqrt{x}\Big|_k^1 = 2\left(\frac{\pi}{4}-0\right) = \frac{\pi}{2}$.

Ex3: $\int_0^1 \frac{dx}{\sqrt{1-x}} = \lim_{k\to 1^-} \int_0^k (1-x)^{-\frac{1}{2}}dx$

$= \lim_{k\to 1^-} -2\sqrt{1-x}\Big|_0^k = \lim_{k\to 1^-} -2\sqrt{1-k} + 2 = +2$

$\therefore$ integral converge to $+2$.

Ex4: $\int_0^2 \frac{dx}{(x-1)^2}$

$= \int_0^1 \frac{dx}{(x-1)^2} + \int_1^2 \frac{dx}{(x-1)^2}$

$= \lim_{k_1\to 1^-} \int_0^{k_1} \frac{dx}{(x-1)^2} + \lim_{k_2\to 1^+} \int_{k_2}^2 \frac{dx}{(x-1)^2}$

$= \lim \frac{-1}{x-1}\Big|_0^{k_1} + \lim \frac{-1}{x-1}\Big|_{k_2}^2$

$= \lim_{k_1\to 1^-}\left(\frac{-1}{k_1-1}-1\right) + \lim_{k_2\to 1^+} -1-\frac{-1}{k_2-1}$

$= +\infty + \infty = +\infty \Rightarrow$ integral diverge

Ex: $\int_0^\infty \frac{dx}{\sinh x + 3\cosh x}$

$= \lim_{L\to+\infty} \int_0^L \frac{2\,dx}{e^x - e^{-x} + 3e^x + 3e^{-x}} = \lim \int_0^L \frac{dx}{2e^x + e^{-x}} \cdot \frac{e^x}{e^x} = \lim \int_0^L \frac{e^x}{2e^{2x}+1}\,dx$

$z = e^x \to = \lim \int \frac{dz}{2z^2+1} \xrightarrow{u=\sqrt{2}z} \lim \int \frac{\frac{1}{\sqrt{2}}du}{u^2+1}$

$= \lim \frac{1}{\sqrt{2}} \tan^{-1}\sqrt{2}z = \lim_{L\to\infty} \frac{1}{\sqrt{2}} \tan^{-1}\sqrt{2}e^x \Big|_0^L = \frac{1}{\sqrt{2}}\left(\frac{\pi}{2} - \tan^{-1}\sqrt{2}\right)$

Ex: $\int_2^\infty \frac{dx}{x^2\sqrt{x^2-4}}$

$= \lim_{k_1\to 2^+} \int_{k_1}^3 \frac{dx}{x^2\sqrt{x^2-4}} + \lim_{k_2\to+\infty} \int_3^{k_2} \frac{dx}{x^2\sqrt{x^2-4}}$

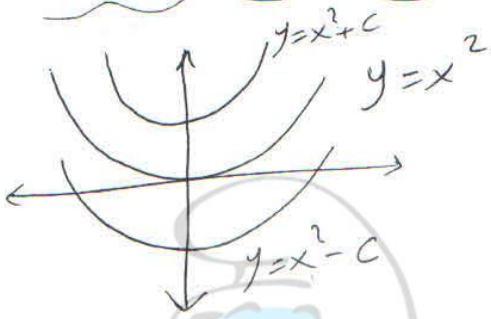
$x = 2\sec\theta$ use trig substitutions.

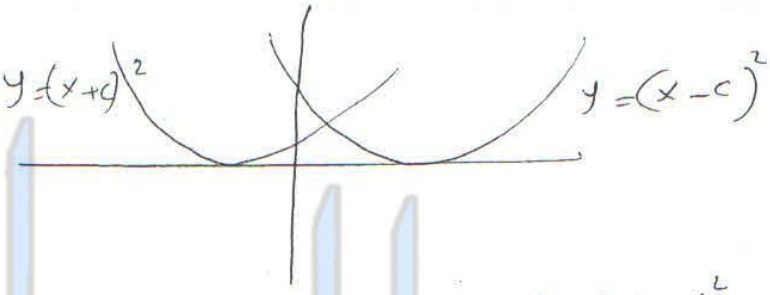
$= \lim_{k_1\to 2^+} \frac{\sqrt{x^2-4}}{4x}\Big|_{k_1}^3 + \lim \frac{\sqrt{x^2-4}}{4x}\Big|_3^{k_2}$

$= \cancel{\Gamma(3)} - \lim_{k_1\to 2^+} \frac{\sqrt{k_1^2-4}}{4k_1} + \lim_{k_2\to+\infty} \frac{\sqrt{k_2^2-4}}{4k_2} - \cancel{\Gamma(3)}$

$= 0 + \frac{1}{4} = \frac{1}{4}$.

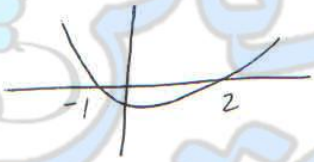
Some important graphs



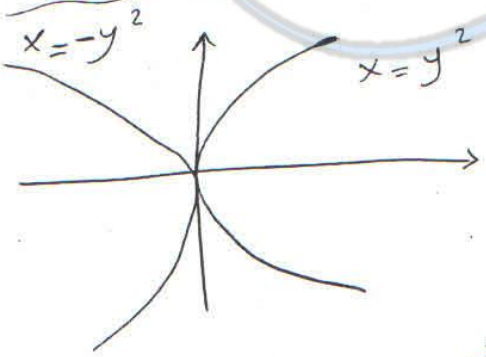


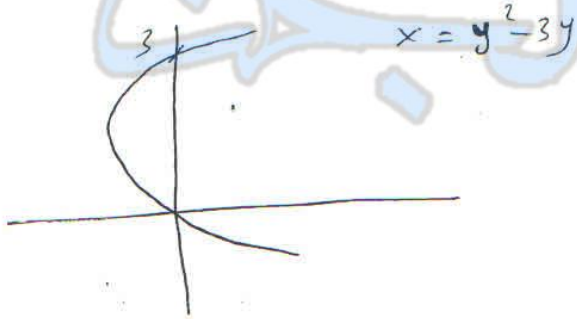


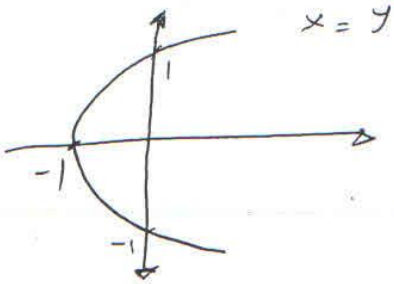


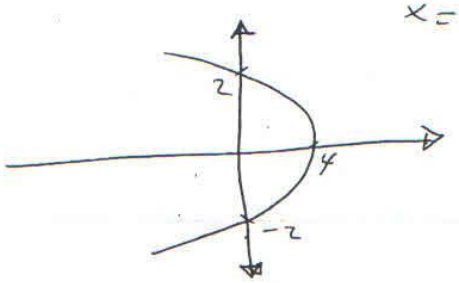


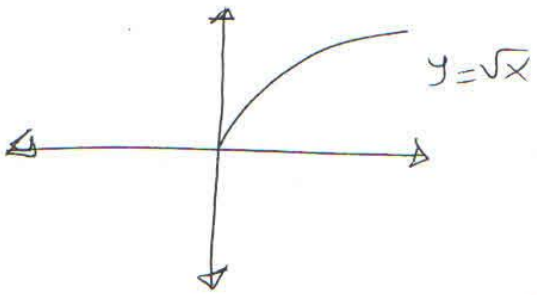
$y=x^2-x-2$
$=(x-2)(x+1)$

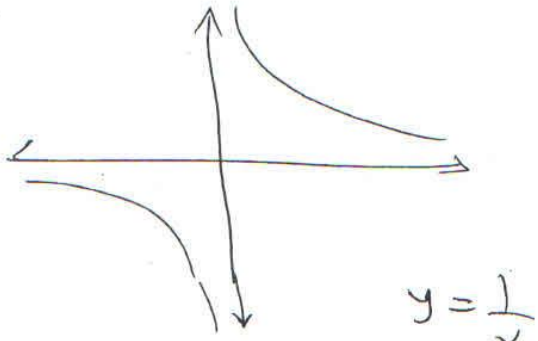


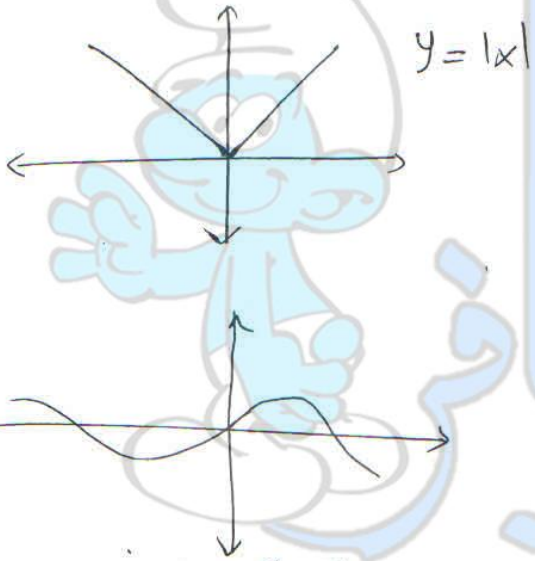


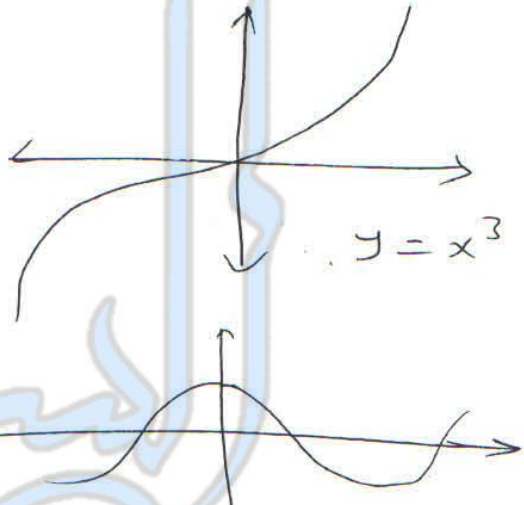


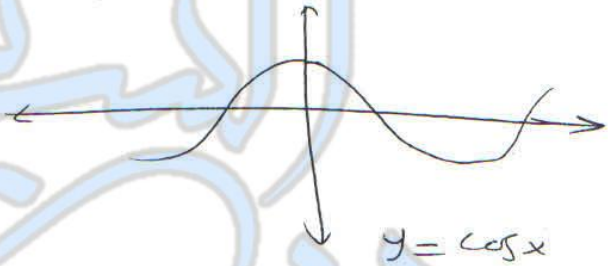


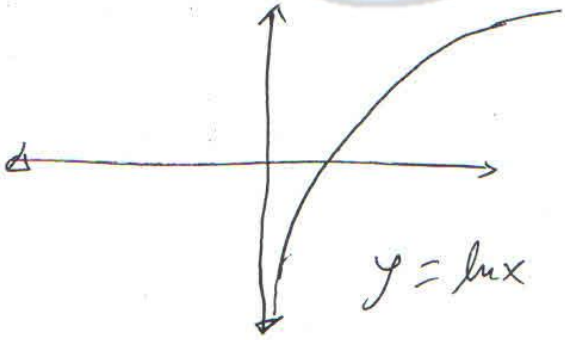


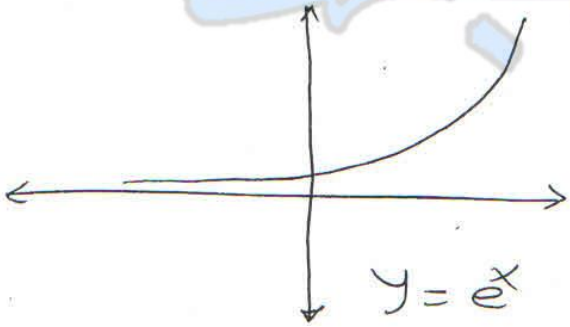


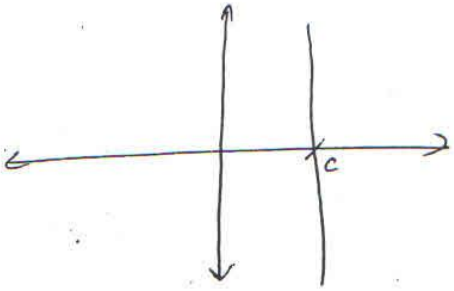






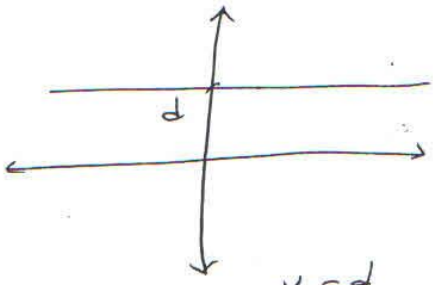






$x=c$

if $c>0$



$y=d$

if $d>0$

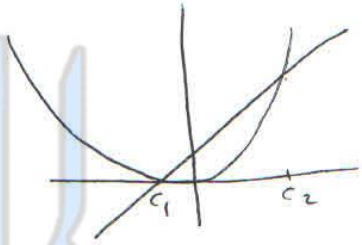
Area between two curves

To find area between $f(x)$ and $g(x)$

1- find intersection between f and g

2- find $I = \int_{c_1}^{c_2} f - g \, dx$

3- $A = |I|$



To find area between f and g with given interval $[a, b]$

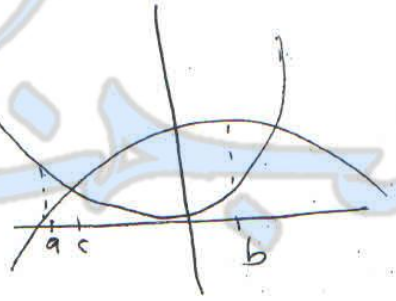
1- find intersection pts that lie on $[a, b]$

2- find $I_1 = \int_a^c f - g \, dx$

$I_2 = \int_c^b f - g \, dx$

and all integrals if more intersections exist.





3- $A = |I_1| + |I_2|$

* Make the same steps if we have x is a function of y.

$x = u(y)$, $x = v(y)$

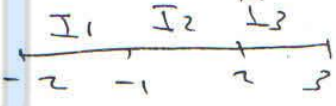
Ex 1: Find enclosed area between $y = x^2$ and $y - x = 6$.

$y = x^2$, $y = x + 6$ $\longrightarrow$ $f = g \rightarrow x^2 = x + 6 \rightarrow x^2 - x - 6 = 0$
$\rightarrow x = -2, 3$

$$I = \int_{-2}^{3} f - g \, dx = \int_{-2}^{3} x^2 - x - 6 \, dx = \frac{-125}{6} \rightarrow A = \frac{125}{6}.$$

Ex 2: Find area of the region between the curves $x^2 + y = 4$ and $y + x = 2$ on $[-2, 3]$

$f = g \longrightarrow y - x^2 = 2 - x \longrightarrow x^2 - x - 2 = 0 \longrightarrow x = -1, 2$



$$I_1 = \int_{-2}^{-1} f - g \, dx = \frac{-11}{6}$$

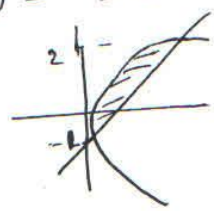
$$I_2 = \int_{-1}^{2} f - g \, dx = \frac{9}{2}, \quad I_3 = \int_{+2}^{3} f - g \, dx = \frac{-11}{6}.$$

$$A = \frac{11}{6} + \frac{9}{2} + \frac{11}{6} = \frac{49}{6}.$$

Ex 3: Find area enclosed between the curves $x - y^2 = 0$ and $y + 2 = x$.

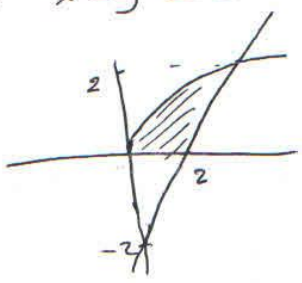
$x = y^2$, $x = y + 2$ $\rightarrow u(y) = v(y) \longrightarrow y = -1, 2$.

$$I = \int_{-1}^{2} (y + 2) - y^2 \, dy = \frac{9}{2} \longrightarrow A = \frac{9}{2}.$$



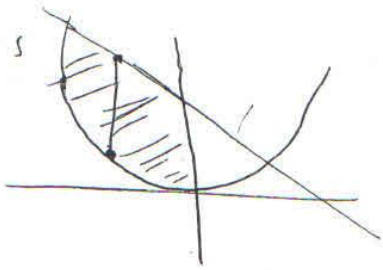
Ex 4: Find area enclosed between $x - y^2 = 0$ and $y + 2 = x$ in the first coordinate

$$A = \int_{0}^{2} y + 2 - y^2 \, dy = \frac{10}{3}.$$



Ex: Find area enclosed between

a) $y = x^2$, $x + y = 2$ and y-axis in the 2nd quadrant

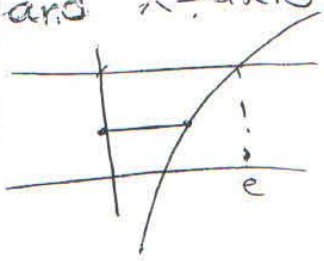


$y = 2 - x$, $y = x^2$, $-2 \le x \le 0$

$x^2 = 2 - x$
$x^2 + x - 2 = 0$
$x = 1, -2$

$$A = \int_{-2}^{0} 2 - x - x^2 \, dx = \frac{10}{3}$$

b) $y = \ln x$, $y = 1$, y-axis and x-axis



$x = e^y$, $x = 0$, $0 \le y \le 1$

$$A = \int_{0}^{1} e^y - 0 \, dy = e - 1$$

c) $y = \sqrt{2 - x}$, $x = -y$ and x-axis



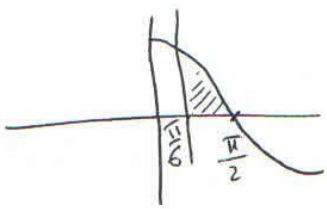
$2 - y^2 = -y$
$y^2 - y - 2 = 0$
$y = 2, -1$

$x = 2 - y^2$, $x = -y$, $0 \le y \le 2$

$$A = \int_{0}^{2} 2 - y^2 - -y \, dy = \frac{10}{3}$$

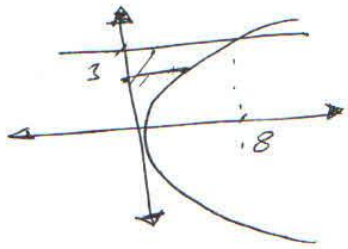
d) $x = \cos^{-1} y$ $x = \frac{\pi}{6}$ and x-axis in 1st quadrant.

$y = \cos x$, $y = 0$, $\frac{\pi}{6} \le x \le \frac{\pi}{2}$



$$A = \int_{\pi/6}^{\pi/2} \cos x \, dx = \sin x \Big|_{\frac{\pi}{6}}^{\frac{\pi}{2}} = 1 - \frac{1}{2} = \frac{1}{2}$$

Ex: Find area enclosed between $x = 2y^2$, $x = 0$ and $y = 3$

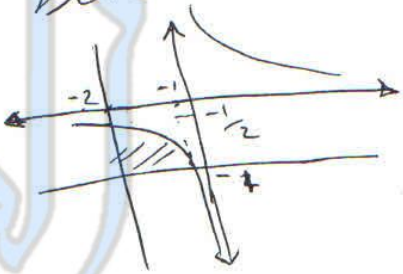


$x = 2y^2$, $x = 0$, $0 \le y \le 3$

$$A = \int_0^3 2y^2 - 0 \, dy = 18.$$

or $$A = \int_0^{18} 3 - \sqrt{\frac{x}{2}} \, dx = 18$$

Ex: Find area enclosed between



$x = \frac{1}{y}$, $x = -2$, $y = -1$

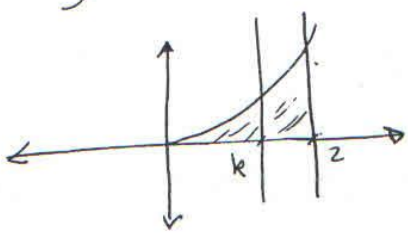
$y = \frac{1}{x}$, $y = -1$, $-2 \le x \le -1$

$$A = \int_{-2}^{-1} \frac{1}{x} - -1 \, dx = 1 - \ln 2.$$

OR $x = \frac{1}{y}$, $x = -1$, $-1 \le y \le -\frac{1}{2}$.

$$A = \int_{-1}^{-\frac{1}{2}} \frac{1}{y} - -1 \, dy = 1 - \ln 2.$$

Ex: Find a vertical line $x = k$ that divides the area enclosed by $x = \sqrt{y}$, $x = 2$ and y-axis into two equal parts



$$\int_0^k x^2 \, dx = \int_k^2 x^2 \, dx \rightarrow k = \sqrt[3]{4}.$$

Ex: Find area enclosed by

$y = \frac{2}{1+x^2}$, $y = |x|$

$\frac{2}{1+x^2} = x$, $x \geq 0 \rightarrow x^3 + x - 2 = 0 \rightarrow x = 1$

$\frac{2}{1+x^2} = -x$, $x \leq 0 \rightarrow x^3 + x + 2 = 0 \rightarrow x = -1$

$A_1 = \int_0^1 \frac{2}{1+x^2} - x \, dx = 2\tan^{-1}x - \frac{x^2}{2} \Big|_0^1 = \frac{2\pi}{4} - \frac{1}{2} - 0 = \frac{\pi - 1}{2}$

$A_2 = \int_{-1}^0 \frac{2}{1+x^2} - -x \, dx = 2\tan^{-1}x + \frac{x^2}{2} \Big|_{-1}^0 = 0 - \left(2\left(\frac{-\pi}{4}\right) + \frac{1}{2}\right) = \frac{\pi - 1}{2}$

$\rightarrow A = \pi - 1$

Ex: Find area enclosed by

$y = \frac{1}{x\sqrt{1-(\ln x)^2}}$ and $y = \frac{2}{x}$

$f(x) = g(x) \rightarrow \frac{1}{x\sqrt{1-(\ln x)^2}} = \frac{2}{x} \rightarrow 1 - (\ln x)^2 = \frac{1}{4} \rightarrow (\ln x)^2 = \frac{3}{4}$

$\rightarrow \ln x = \pm \frac{\sqrt{3}}{2} \rightarrow x = e^{\pm \sqrt{3}/2}$

$$A = \int_{e^{-\sqrt{3}/2}}^{e^{\sqrt{3}/2}} \frac{2}{x} - \frac{1}{x\sqrt{1-(\ln x)^2}} \, dx$$

$$= 2\ln|x| - \sin^{-1}(\ln x) \Big|_{e^{-\sqrt{3}/2}}^{e^{\sqrt{3}/2}}$$

$$= \left(2\frac{\sqrt{3}}{2} - \frac{\pi}{3}\right) - \left(2\left(\frac{-\sqrt{3}}{2}\right) - \frac{-\pi}{3}\right) = 2\sqrt{3} - \frac{2\pi}{3}$$

$\int \frac{1}{x\sqrt{1-(\ln x)^2}} \, dx$

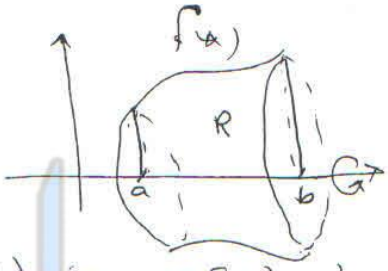
$z = \ln x$, $dz = \frac{1}{x} dx$

$= \int \frac{dz}{\sqrt{1-z^2}} = \sin^{-1} z$

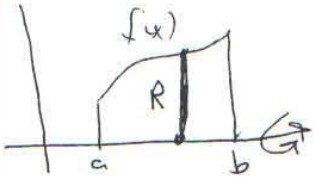
Volume of revolution

[1] Disk method

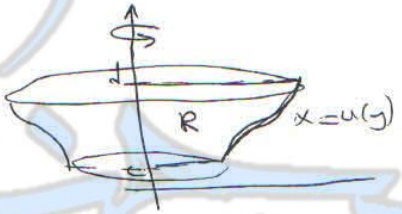
* When region R between $y = f(x)$ and x-axis, $a \le x \le b$ revolved about x-axis we obtain a solid has a volume, the volume of this solid is

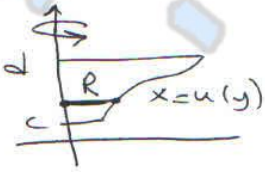


$$V = \pi \int_a^b f^2(x)\, dx$$



* When a region R between $x = u(y)$ and y-axis $c \le y \le d$ is revolved about y-axis the volume of this solid is

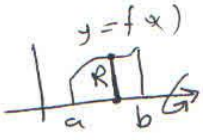




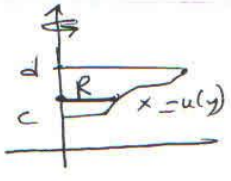
$$V = \pi \int_c^d u^2(y)\, dy$$

* Summary

(1) If R revolved about x-axis $y = f(x)$, $y = 0$ $a \le x \le b \rightarrow V = \pi \int_a^b f^2(x)\, dx$



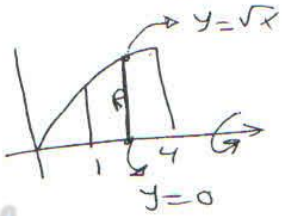
(2) If R revolved about y-axis $x = u(y)$, $x = 0$, $c \le y \le d$, $\rightarrow V = \pi \int_c^d u^2(y)\, dy$



** Cross section in disk method perpendicular to axis of revolution

Examples

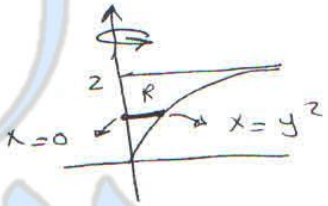
① Find volume of solid obtained when region between $y=\sqrt{x}$ and x-axis over $[1,4]$ is revolved about x-axis



$y=\sqrt{x}$, $y=0$, $1 \le x \le 4$

$$V = \pi \int_1^4 (\sqrt{x})^2 \, dx = \frac{15\pi}{2}$$

② Find volume of solid generated when region enclosed by $y=\sqrt{x}$, $y=2$ and $x=0$ is revolved about y-axis



$x=y^2$, $x=0$, $0 \le y \le 2$

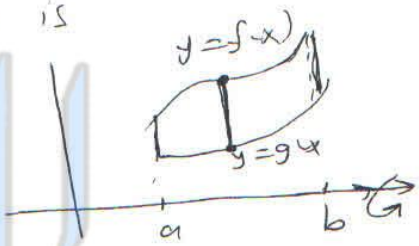
$$V = \pi \int_0^2 (y^2)^2 \, dy = \frac{32\pi}{5}$$

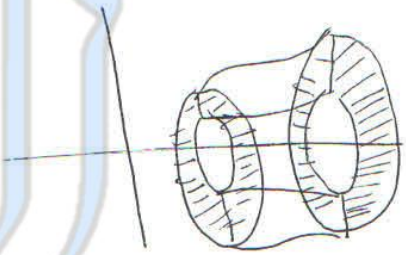
2 Washer method

* When region R between $y = f(x)$ and $y = g(x)$ $a \le x \le b$ is revolved about x-axis,

the volume of solid obtained is

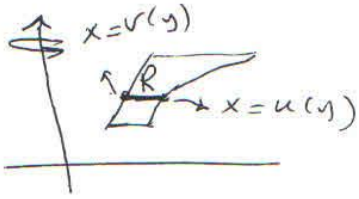
$$V = \pi \int_a^b f^2(x) - g^2(x)\, dx$$

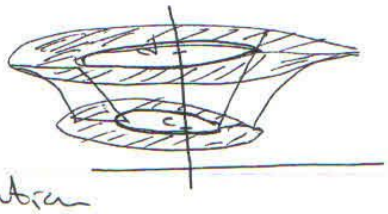




* When a region R between $x = u(y)$ and $x = v(y)$, $c \le y \le d$ is revolved about y-axis, the volume of solid is

$$V = \pi \int_c^d u^2(y) - v^2(y)\, dy$$





* Cross section in washer method is perpendicular to axis of revolution

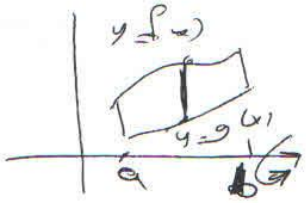
* If the cross section vertical | we write $y = f(x)$, $y = g(x)$

If the cross section horizontal — we write $x = u(y)$, $x = v(y)$

Summary of washer method

* If R revolved about x-axis

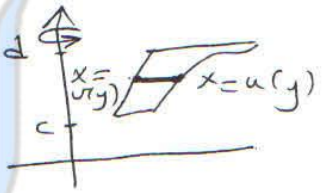
$y = f(x)$, $y = g(x)$, $a \le x \le b$



$$V = \pi \int_a^b f^2(x) - g^2(x)\, dx$$

* If R is revolved about y-axis

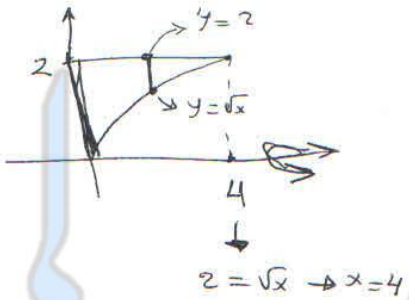
$x = u(y)$, $x = v(y)$, $c \le y \le d$



$$V = \pi \int_c^d u^2(y) - v^2(y)\, dy$$

Examples

① Find volume of solid generated when region between $y=\sqrt{x}$, $y=2$ and $x=0$ is revolved about x-axis



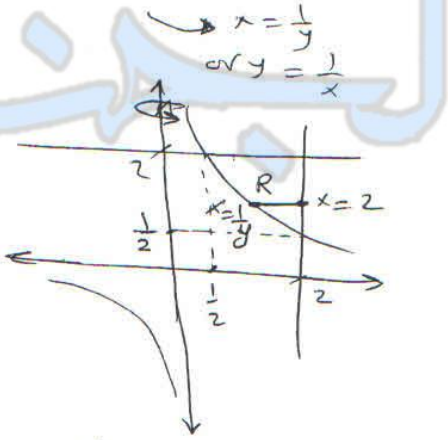
$y=2$, $y=\sqrt{x}$, $0 \le x \le 4$

$$V = \pi \int_0^4 2^2 - (\sqrt{x})^2 \, dx = 8\pi$$

② Find volume of solid generated when region between $y=2$, $x=2$ and $xy=1$ is

a) revolved about y-axis

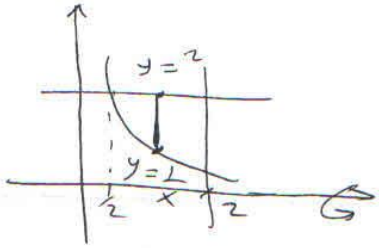
b) " " x-axis



a) $x=2$, $x=\frac{1}{y}$, $\frac{1}{2} \le y \le 2$

$$V = \pi \int_{\frac{1}{2}}^{2} 4 - \frac{1}{y^2} \, dy = \frac{9}{2}\pi$$

b) $y=2$, $y=\frac{1}{x}$, $\frac{1}{2} \le x \le 2$

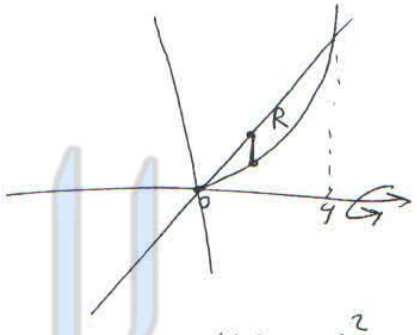


$$V = \pi \int_{\frac{1}{2}}^{2} 4 - \frac{1}{x^2} \, dx = \frac{9}{2}\pi$$

(3) Region enclosed between $x=\sqrt{y}$ and $\frac{y}{x}=4$ is revolved about x-axis find volume of the solid generated.

$x=\sqrt{y}$ or $y=x^2$

$\frac{y}{x}=4$ or $y=4x$



$R: y=4x,\ y=x^2,\ 0\le x\le 4$

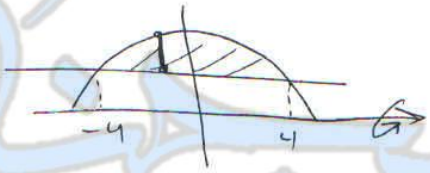
$4x=x^2$

$x^2-4x=0$

$x=0,4$

$$V=\pi\int_0^4 (4x)^2-(x^2)^2\,dx = \frac{2048\pi}{15}$$

(4) Region enclosed by $y=\sqrt{25-x^2}$, $y=3$ revolved about x-axis



$y=\sqrt{25-x^2},\ y=3,\ -4\le x\le 4$

$\sqrt{25-x^2}=3$

$25-x^2=9$

$x^2=16$

$x=\pm 4$

$$V=\pi\int_{-4}^{4}(25-x^2)-9\,dx$$

$$=16x-\frac{x^3}{3}\Big|_{-4}^{4}=\left(\underbrace{16(4)}_{4^3}-\frac{4^3}{3}\right)-\left(\underbrace{-16(4)}_{-4^3}+\frac{4^3}{3}\right)$$

$$=2\cdot 4^3-2\frac{4^3}{3}=4^3\left(2-\frac{2}{3}\right)=4^3\cdot\frac{4}{3}=\frac{4^4}{3}$$

(5) Region enclosed by $x=y^2$ and $y=x-2$ revolved about y-axis

$x=y+2,\ x=y^2,\ -1\le y\le 2$



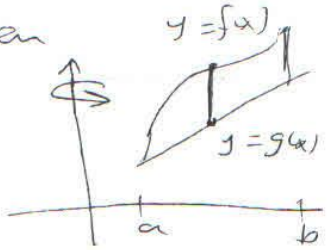
$y^2=y+2 \rightarrow y^2-y-2=0$

$\rightarrow y=2,-1$

$$V=\pi\int_{-1}^{2}(y+2)^2-y^4\,dy=\frac{72}{5}\pi$$

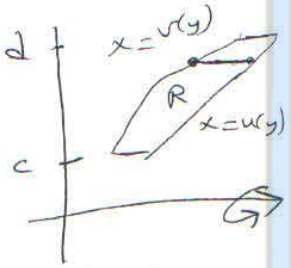
3 Cylindrical shell method

* When a region R between $y = f(x)$ and $y = g(x)$, $a \le x \le b$ is revolved about y-axis the volume of solid is



$$V = 2\pi \int_a^b x\,(f(x) - g(x))\,dx$$

* When a region R between $x = u(y)$, $x = v(y)$, $c \le y \le d$ is revolved about x-axis the volume of solid is



$$V = 2\pi \int_c^d y\,(u(y) - v(y))\,dy$$

In cylindrical shell method cross section parallel to axis of revolution

* Summary

(1) if R revolved about x-axis

put $x = u(y)$, $x = v(y)$ $c \le y \le d$

$$V = 2\pi \int_c^d y\,(u(y) - v(y))\,dy$$

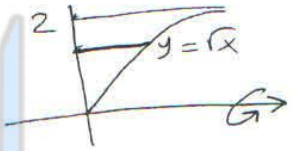
(2) if R revolved about y-axis

put $y = f(x)$, $y = g(x)$, $a \le x \le b$

$$V = 2\pi \int_a^b x\,(f(x) - g(x))\,dx$$

Examples

① Find volume of solid generated when region between $y=\sqrt{x}$, $y=2$ and $x=0$ is revolved about x-axis (use cylindrical method)

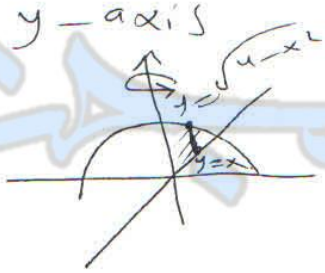


revolved about x-axis

$x=y^2$, $x=0$, $0\le y\le 2$

$$V=2\pi\int_0^2 y(y^2-0)\,dy = 2\pi\left(\frac{y^4}{4}\Big|_0^2\right) = 8\pi .$$

② Region between $y=\sqrt{4-x^2}$, $y=x$ in the 1st quadrant revolved about y-axis find volume of solid



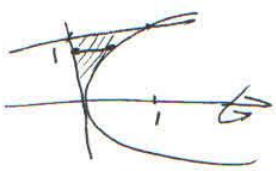
revolved about y-axis

$y=\sqrt{4-x^2}$, $y=x$, $0\le x\le \sqrt{2}$

$$V=2\pi\int_0^{\sqrt{2}} x\left(\sqrt{4-x^2}-x\right)dx = 2\pi\left(\frac{8-2\sqrt{2}}{3}\right)$$

③ Region between $x=y^2$, $y=1$ and $x=0$ revolved about (x-axis)

$$V=2\pi\int_0^1 y(y^2-0)\,dy = \frac{\pi}{2} .$$

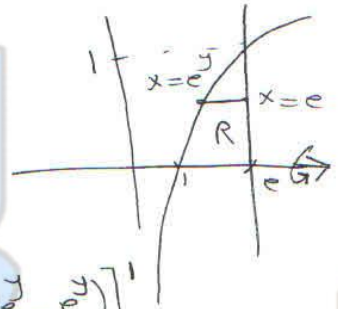


$x=y^2$, $x=0$

$0\le y\le 1$

Examples:

(1) Find volume of solid when region R between $y = \ln x$, $x = e$ and x-axis is revolved about x-axis



revolved about x-axis

$x = e$, $x = e^y$, $0 \le y \le 1$

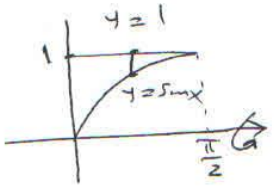
$$V = 2\pi \int_0^1 y(e - e^y)\,dy = 2\pi \left[\frac{ey^2}{2} - (ye^y - e^y) \right]_0^1$$

$$= 2\pi \left[\left(\frac{e}{2} - 0 \right) - (1) \right] = 2\pi \left(\frac{e-2}{2} \right) = \pi(e-2)$$

(2) Region between $y = \cos x^2$, $x = 0$, $x = \frac{\sqrt{\pi}}{2}$, $y = 0$ revolved about y-axis

$$V = 2\pi \int_0^{\frac{\sqrt{\pi}}{2}} x(\cos x^2 - 0)\,dx = \frac{2\pi}{2} \sin x^2 \Big|_0^{\frac{\sqrt{\pi}}{2}} = \pi\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}$$

(3) region between $x = \sin^{-1} y$, $y = 1$ and y-axis revolved about x-axis



$$V = \pi \int_0^{\frac{\pi}{2}} 1^2 - (\sin x)^2\,dx = \pi \int 1 - \left(\frac{1 - \cos 2x}{2}\right) dx$$

$$= \pi \left[\frac{x}{2} + \frac{\sin 2x}{4} \Big|_0^{\frac{\pi}{2}} \right]$$

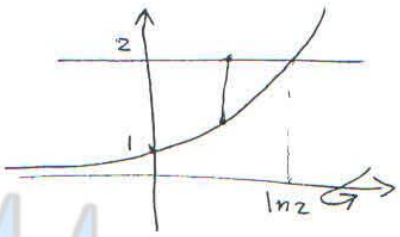
$$= \pi \left[\left(\frac{\pi}{4} + 0 \right) - (0) \right] = \frac{\pi^2}{4}.$$

$x = \sin^{-1} y \iff y = \sin x$

(4) Region enclosed by $x = \ln y$, $y = 2$ and y-axis revolved about ~~x~~-axis

a) by Washer method

b) by Cylindrical shell method

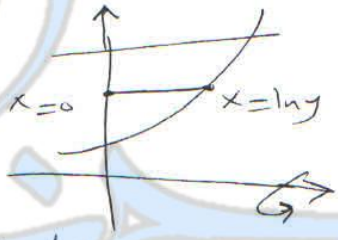


a) revolve about x-axis

$y = e^x$, $y = 2$, $0 \le x \le \ln 2$. $\quad x = \ln y \leftrightarrow y = e^x$

$$V = \pi \int_0^{\ln 2} 4 - e^{2x}\, dx = \pi \left. 4x - \frac{e^{2x}}{2} \right|_0^{\ln 2} = \frac{\pi}{2}[8\ln 2 - 3]$$

b) revolve about x-axis



$x = \ln y$, $x = 0$, $1 \le y \le 2$

$$V = 2\pi \int_1^2 y(\ln y - 0)\, dy = 2\pi \int_1^2 y \ln y\, dy$$

$$= 2\pi \left[\frac{y^2}{2}\ln y - \frac{y^2}{4} \right|_1^2 = \frac{\pi}{2}[8\ln 2 - 3].$$

Exercise:
Let V_x and V_y be volumes of solids result when the region enclosed by $y = \frac{1}{x}$, $y = 0$, $x = \frac{1}{2}$ and $x = b$ $(b > \frac{1}{2})$ is revolved about x-axis and y-axis respectively. Find b for which $V_x = V_y$.

Area of surface of revolution:

*If the curve $y=f(x)$, $a\le x\le b$ is revolved about x-axis, then the surface area S of the surface of revolution that is generated by revolving $f(x)$, $a\le x\le b$ about x-axis is

$$S=\int_a^b 2\pi f(x)\sqrt{1+(f'(x))^2}\,dx=\int_a^b 2\pi y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

* If the curve $x=g(y)$, $c\le y\le d$ is revolved about y-axis, then the surface area of surface of revolution that is generated by revolving $g(y)$, $c\le y\le d$ about y-axis is

$$S=\int_c^d 2\pi g(y)\sqrt{1+(g'(y))^2}\,dy=\int_c^d 2\pi x\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$$

Examples

(1) Find area surface that is generated by revolving the portion of the curve $y=x^3$, $0\le x\le 1$ about x-axis

$$S=2\pi\int_0^1 x^3\sqrt{1+(3x^2)^2}\,dx=\pi\int_0^1 x^3\sqrt{1+9x^4}\,dx \qquad z=1+9x^4,\ dz=36x^3dx$$

$$=\frac{2\pi}{36}\int z^{\frac12}dz=\frac{1}{18}\cdot\frac{\pi_2 z^{3/2}}{3}\Big|_1^{10}=\frac{1}{27}\left((\sqrt{10})^3-1\right)$$

② Find area of the surface by revolving $y = x^2$, $1 \le x \le 2$ about y-axis.

revolve about y-axis $\longrightarrow x = \sqrt{y}$, $1 \le y \le 4$

$$S = 2\pi \int_1^4 \sqrt{y}\sqrt{1+\frac{1}{4y}}\,dy = 2\pi \int_1^4 \frac{\sqrt{4y+1}}{2} = \frac{\pi}{6}\left(17^{3/2} - 5^{3/2}\right)$$

③ $y = \sqrt{4-x^2}$, $-1 \le x \le 1$ revolve about x-axis

$$S = 2\pi \int \sqrt{4-x^2}\sqrt{1+\left(\frac{-2x}{2\sqrt{4-x^2}}\right)^2}\,dx = 2\pi \int \sqrt{4}\,dx = 8\pi$$

④ $y = 1 - \frac{x^2}{4}$, $2 \le x \le 2\sqrt{2}$, revolve y-axis.

$x = 2\sqrt{1-y}$ $\quad x=2 \to y=0$, $x=2\sqrt{2} \to y=-1$

$$S = 2\pi \int 2\sqrt{1-y}\sqrt{1+\left(\frac{-1}{2\sqrt{1-y}}\right)^2}\,dy = 2\pi \int 2\sqrt{1-y}\sqrt{1+\frac{1}{1-y}}\,dy$$

$$= 2\pi \int 2\sqrt{1-y}\,\frac{\sqrt{2-y}}{\sqrt{1-y}} = 4\pi \cdot \frac{2}{3}(2-y)^{3/2}\Big|_{-1}^{0} = \frac{8\pi}{3}\left(3\sqrt{3} - 2\sqrt{2}\right)$$

⑤ $x - 9y = 1$, $0 \le y \le 2$ about y-axis

$$S = 2\pi \int (1+9y)\sqrt{1+9^2}\,dy = 2\pi\sqrt{82}\,\frac{(1+9y)^2}{18}\Big|_0^2$$

$$= 40\sqrt{82}\,\pi$$

Length of a plane of a curve

If $f(x)$ is smooth f= on $[a,b]$, then the arc length L of the curve $y=f(x)$ from $x=a$ to $x=b$ is

$$L=\int_a^b \sqrt{1+(f'(x))^2}\,dx = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$



If we write the f= in the form

$x=g(y)$, $c\le y\le d$

$$L=\int_c^d \sqrt{1+(g'(y))^2}\,dy = \int_c^d \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$$

* Writing the f= in terms of x or in terms of y depend on which is easier to integrate.

Examples:

Find the arc length for the following curves

① $y^2=x^3$ from $(1,1)$ to $(2,2\sqrt{2})$

$y=x^{3/2} \Rightarrow L=\int_1^2 \sqrt{1+\left(\frac{3}{2}x^{1/2}\right)^2}\,dx = \frac{1}{27}\left[22\sqrt{22}-13\sqrt{13}\right]$

② $y = \frac{1}{8}x^2 - \ln x \quad , \quad 1 \le x \le e$

$\frac{dy}{dx} = \frac{1}{4}x - \frac{1}{x} \rightarrow \left(\frac{dy}{dx}\right)^2 = \left(\frac{x^2}{16} - \frac{1}{2} + \frac{1}{x^2}\right) \rightarrow 1 + \left(\frac{dy}{dx}\right)^2 =$

$1 + \frac{x^2}{16} - \frac{1}{2} + \frac{1}{x^2} = \frac{x^2}{16} + \frac{1}{2} + \frac{1}{x^2} = \left(\frac{x}{4} + \frac{1}{x}\right)^2$

$\therefore L = \int_1^e \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int \frac{x}{4} + \frac{1}{x}\, dx = \frac{x^2}{8} + \ln|x| \Big|_1^e$

$= \frac{e^2}{8} + 1 - \frac{1}{8} = \frac{e^2 - 1}{8} + 1 = \frac{e^2 + 7}{8}$.

③ $x = \frac{y^3}{24} + \frac{2}{y} \quad , \quad 2 \le y \le 4$

$\frac{dx}{dy} = \frac{y^2}{8} - \frac{2}{y^2} \rightarrow \left(\frac{dx}{dy}\right)^2 = \frac{y^4}{64} - \frac{1}{2} + \frac{4}{y^4}$

$\left(\frac{dx}{dy}\right)^2 + 1 = \frac{y^4}{64} + \frac{1}{2} + \frac{4}{y^4} = \left(\frac{y^2}{8} + \frac{2}{y^2}\right)^2$

$\therefore L = \int_2^4 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \int \frac{y^2}{8} + \frac{2}{y^2}\, dy = \frac{11}{2}$.

④ $x = \ln(\sec y) \qquad 0 \le y \le \frac{\pi}{4}$

$L = \int_0^{\frac{\pi}{4}} \sqrt{1 + \left(\frac{\sec y \tan y}{\sec y}\right)^2}\, dy = \int \sqrt{\sec^2 y}\, dy = \int \sec y\, dy$

$= \ln|\sec y + \tan y| \Big|_0^{\frac{\pi}{4}} = \ln(\sqrt{2} + 1)$

⑤ $y = x^{2/3} \quad , \quad 1 \le x \le 8$

$\rightarrow x = y^{3/2} \qquad 1 \le y \le 4$

$L = \frac{1}{27}\left(80\sqrt{10} - 13\sqrt{13}\right)$.

Sequences

Def:
Sequence is an unending succession of numbers, called terms, written as

$$\{a_1, a_2, a_3, \dots\} = \{a_n\}_{n=1}^{\infty} \leftarrow \text{general term}$$

Sequence → converge if $\lim_{n\to\infty} a_n = L$
Sequence → diverge if $\lim_{n\to\infty} a_n = \pm\infty$

Examples: Determine whether the seq. converge or diverge

① $\left\{\frac{n}{2n+1}\right\}_{n=1}^{\infty}$, $\lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2}$ → seq. converge to $\frac{1}{2}$

② $\{8-2n\}_{n=1}^{\infty}$, $\lim_{n\to\infty} 8-2n = -\infty$ → seq. diverge

③ $\{\sqrt[n]{n}\}_{n=1}^{\infty}$, $\lim_{n\to\infty} n^{\frac{1}{n}} = \infty^0 \xrightarrow[\text{L'Hôpital}]{\text{by}} = 1$ → converge to 1

④ $\left\{\frac{1}{\sqrt{\pi}}, \frac{4}{\sqrt[3]{\pi}}, \frac{9}{\sqrt[4]{\pi}}, \frac{16}{\sqrt[5]{\pi}}, \dots\right\}$

$= \left\{\frac{n^2}{\pi^{1/n}}\right\}_{n=1}^{\infty}$, $\lim_{n\to\infty} \frac{n^2}{\pi^{1/n}} = \frac{+\infty}{1} = +\infty$ → diverge

⑤ $\left\{n \sin\frac{\pi}{n}\right\}_{n=1}^{\infty}$, $\lim n\sin\left(\frac{\pi}{n}\right) = \lim_{n\to\infty} \frac{\sin(\pi/n)}{1/n} = \lim_{\theta\to 0} \frac{\sin\pi\theta}{\theta} = \pi$
→ conv. to π

⑥ $\left\{\left(\frac{n-2}{n}\right)^n\right\}_{n=1}^{\infty}$, $\lim \left(\frac{n-2}{n}\right)^n = \lim \left(1-\frac{2}{n}\right)^n = e^{-2}$
→ conv to e^{-2}

⑦ $\left\{\frac{4^n}{\pi^n}\right\}_{n=1}^{\infty}$, $\lim \left(\frac{4}{\pi}\right)^n = +\infty$ since $\frac{4}{\pi} > 1$
→ seq. diverge

⑧ $\left\{\left(\frac{n+2}{n+3}\right)^n\right\}$, $\lim \left(\frac{n+2}{n+3}\right)^n = \lim \frac{(n+2)^n}{(n+3)^n} = \lim_{n\to\infty} \frac{(1+2/n)^n}{(1+3/n)^n}$

$= \frac{e^2}{e^3} = e^{-1}$ → conv. to e^{-1}

For the sequence of the form

$\{(-1)^n a_n\}_{n=1}^{\infty}$ or $\{\cos(n\pi)\, a_n\}_{n=1}^{\infty}$

Take $\{a_n\}_{n=1}^{\infty}$ find $\lim_{n\to\infty} a_n = L$

if $L = 0 \rightarrow \{(-1)^n a_n\}$ conv. to 0.

if $L \neq 0 \rightarrow \{(-1)^n a_n\}$ diverge

Ex:

① $\left\{\frac{(-1)^n}{n^2}\right\}_{n=1}^{\infty}$ $\lim_{n\to\infty} \frac{1}{n^2} = 0 \rightarrow$ seq. conv. to 0.

② $\left\{\frac{\cos(n\pi)(n+1)(n+2)}{2n^2}\right\}_{n=1}^{\infty} = \left\{\frac{(-1)^n (n+1)(n+2)}{2n^2}\right\}_{n=1}^{\infty}$

$\lim \frac{(n+1)(n+2)}{2n^2} = \frac{1}{2} \rightarrow$ seq. diverge.

③ $\left\{1, \frac{-1}{3}, \frac{1}{9}, \frac{-1}{27}, \cdots\right\} = \left\{\frac{(-1)^{n+1}}{3^{n-1}}\right\}_{n=1}^{\infty}$

$\lim_{n\to\infty} \frac{1}{3^{n-1}} = 0 \rightarrow$ seq. conv. to 0.

Monotone Sequences

Any sequence is said to be monotone seq. if this seq. increas. or decreas.

If $\{a_n\}_{n=1}^{\infty}$ satisfy that

① $a_{n+1} > a_n$ or $\frac{a_{n+1}}{a_n} > 1$ or $f'(n) = (a_n)' > 0 \Rightarrow \{a_n\}$ strictly increasing

② $a_{n+1} \geq a_n$ or $\frac{a_{n+1}}{a_n} \geq 1$ or $f'(n) = (a_n)' \geq 0 \Rightarrow \{a_n\}$ increasing

③ $a_{n+1} < a_n$ or $\frac{a_{n+1}}{a_n} < 1$ or $f'(n) = (a_n)' < 0 \Rightarrow \{a_n\}$ strictly decreasing

④ $a_{n+1} \leq a_n$ or $\frac{a_{n+1}}{a_n} \leq 1$ or $f'(n) = (a_n)' \leq 0 \Rightarrow \{a_n\}$ decreasing.

Ex:

① $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\right\} = \left\{\frac{n}{n+1}\right\}$

$\frac{a_{n+1}}{a_n} = \frac{n+1}{n+2} \cdot \frac{n+1}{n} = \frac{n^2+2n+1}{n^2+2n} > 1$ → seq. strictly increasing

or $f' = \frac{1}{(n+1)^2} > 0$

② $\left\{\frac{10^n}{(2n)!}\right\}_{n=1}^{\infty}$

$\frac{a_{n+1}}{a_n} = \frac{10^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{10^n} = \frac{10}{(2n+2)(2n+1)} < 1$ → seq. strictly decreasing

$\left\{n=1 \quad \frac{10}{12} < 1\right\}$

③ $\left\{\frac{2^n}{1+2^n}\right\}_{n=1}^{\infty}$

$f'(n) = \frac{2^n \ln 2 (1+2^n) - 2^n \ln 2 \cdot 2^n}{(1+2^n)^2} = \frac{2^n \ln 2}{(1+2^n)^2} > 0$

→ strictly increasing

* If the seq. can be written as union of odd terms and even terms.

$$\{u_n\}_{n=1}^{\infty} = \underbrace{\{a_n\}_{n=1}^{\infty}}_{\text{odd terms}} \cup \underbrace{\{b_n\}_{n=1}^{\infty}}_{\text{even terms}}$$

if $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = L \Rightarrow \{u_n\}_{n=1}^{\infty}$ conv. to L

otherwise diverge.

Ex:

① $\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{9}, \frac{1}{8}, \cdots\} = \{\frac{1}{2^n}\} \cup \{\frac{1}{3^n}\}$

$\lim_{n\to\infty} \frac{1}{2^n} = 0$, $\lim_{n\to\infty} \frac{1}{3^n} = 0$ → seq. conv. to 0.

② $\{1, \frac{1}{2}, \frac{2}{3}, \frac{1}{2}, \frac{3}{5}, \frac{1}{2}, \frac{4}{7}, \frac{1}{2}, \cdots\}$

$= \{1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \cdots\} \cup \{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \cdots\}$

$= \{\frac{n}{2n-1}\}_{n=1}^{\infty} \cup \{\frac{1}{2}\}_{n=1}^{\infty}$

$\lim \frac{n}{2n-1} = \frac{1}{2}$, $\lim \frac{1}{2} = \frac{1}{2}$ → seq. conv. to $\frac{1}{2}$

③ $\{2, \frac{1}{2}, 3, \frac{1}{4}, 4, \frac{1}{8}, \cdots\} = \{n+1\}_{n=1}^{\infty} \cup \{\frac{1}{2^n}\}_{n=1}^{\infty}$

$\lim_{n\to\infty} (n+1) = \infty$

→ seq. diverge.

Infinite Series

Def:

An infinite series is an expression that can be written in the form

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \cdots$$

$u_1, u_2, \ldots$ are called terms.

Any series may converge or diverge

if the series converge to a number L, then L = Sum of the series.

Sums of infinite series

[1] Series of nth partial sum

For the series $\sum_{k=1}^{\infty} u_k$

take the sequence $\{S_n\}_{n=1}^{\infty}$ where $S_n = u_1 + u_2 + \cdots + u_n$

* If the seq. conv. ($\lim_{n\to\infty} S_n = S$), then the series conv. and $\sum_{k=1}^{\infty} u_k = S$

* If the seq. diverge ($\lim_{n\to\infty} S_n = \infty$), then the series diverge.

Ex: Determine whether the series $\sum_{k=1}^{\infty}(-1)^{k+1}$ = $1-1+1-1+\cdots$ converge or diverge.

$S_1 = u_1 = 1$, $S_2 = u_1+u_2 = 0$, $S_3 = 1$, $S_4 = 0$, ...

i seq. $\{1, 0, 1, 0, \ldots\} = \{1, 1, \ldots\} \cup \{0, 0, \ldots\}$

odd terms conv. to 1; even terms conv. to 0

So seq. div. $\Rightarrow$ Series diverge

Ex: $\sum_{k=1}^{\infty} \frac{3}{10^k}$

seq. $\{0.3, 0.33, 0.333, \ldots\}$

$S_n = \frac{3}{10} + \frac{3}{10^2} + \cdots + \frac{3}{10^n}$ ... ①

$\frac{1}{10} S_n = \frac{3}{10^2} + \frac{3}{10^3} + \cdots + \frac{3}{10^{n-1}}$ ... ②

① - ② $\Rightarrow \frac{9}{10} S_n = \frac{3}{10} - \frac{3}{10^n} \Rightarrow S_n = \frac{1}{3}\left(1 - \frac{1}{10^n}\right)$ $\lim_{n\to\infty} S_n = \frac{1}{3}$

seq. conv. to $\frac{1}{3}$ $\Rightarrow$ Series conv. to $\frac{1}{3}$.

[2] Geometric Series

$$\sum u_k = \sum_{k=0}^{\infty} ar^k = \underbrace{a}_{u_1} + \underbrace{ar}_{u_2} + \underbrace{ar^2}_{u_3} + \cdots$$

is a geometric series if

$\frac{u_{k+1}}{u_k} = r$ for $k \geq 1$ where a = 1'st term

* geometric series conv. if $|r| < 1$ and Sum $= \frac{a}{1-r}$

" " div. if $|r| \geq 1$

Examples:

Determine if the series conv. or div.

(1) $1 + 2 + 4 + 8 + \cdots = \sum_{k=0}^{\infty} 2^k$, $r = 2 > 1 \rightarrow$ div.

(2) $\frac{1}{3} - \frac{1}{9} + \frac{1}{27} - \frac{1}{81} + \cdots = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{3^k}$, $r = -\frac{1}{3}$ $|r| < 1$

∴ series conv. $a = \frac{1}{3} \rightarrow$ Sum $= \frac{a}{1-r} = \frac{1/3}{1-(-\frac{1}{3})} = \frac{1}{4}$.

(3) $-1 + 1 - 1 + \cdots = \sum_{k=1}^{\infty} (-1)^k$ $r = -1$ $|r| \geq 1 \rightarrow$ div.

(4) $\sum_{k=1}^{\infty} 3^{2k} 5^{1-k}$, $r = \frac{9}{5} \rightarrow$ div.

(5) $\sum_{k=1}^{\infty} \frac{(-1)^k 4^{k+2}}{7^{k-1}} = \frac{-4^3}{7^0} + \frac{4^4}{7^1} - \frac{4^5}{7^2} + \cdots$ $r = -\frac{4}{7}$, $|r| < 1 \rightarrow$ Conv.

$a = -4^3 \rightarrow$ Sum $= \frac{-4^3}{1+\frac{4}{7}} = -4^3 \cdot \frac{7}{11}$ ∴

(6) $\sum_{k=2}^{\infty} \frac{1}{2^k} - \frac{1}{2^{k+1}}$

$\sum_{k=2}^{\infty} \frac{1}{2^k}$, $r = \frac{1}{2}$, $Sum = \frac{\frac{1}{4}}{1-\frac{1}{2}} = \frac{1}{2}$.

$\sum_{k=2}^{\infty} \frac{1}{2^{k+1}}$, $r = \frac{1}{2}$, $Sum = \frac{\frac{1}{8}}{1-\frac{1}{2}} = \frac{1}{4}$

$\therefore \sum_{k=2}^{\infty} \frac{1}{2^k} - \frac{1}{2^{k+1}} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

(7) $\sum_{k=5}^{\infty} \left(\frac{e}{\pi}\right)^{k-1}$, $r = \frac{e}{\pi} = \frac{2.71}{3.14} < 1 \rightarrow$ conv.

$Sum = \frac{\left(\frac{e}{\pi}\right)^4}{1-\frac{e}{\pi}}$

(8) $\sum_{k=1}^{\infty} \cos k\pi = \sum_{k=1}^{\infty} (-1)^k$ $r = -1$ div.

(9) $0.3333 = 0.3 + 0.03 + 0.003 + \cdots = \frac{3}{10} + \frac{3}{100} + \frac{3}{10^3} + \cdots$

$r = \frac{1}{10}$ conv. $\rightarrow$ $Sum = \frac{3/10}{1-\frac{1}{10}} = \frac{1}{3}$.

(10) $\sum_{k=1}^{\infty} (\ln 2)^{1-k}$

$= \frac{1}{1} + \frac{1}{\ln 2} + \frac{1}{(\ln 2)^2} + \cdots$

$r = \frac{1}{\ln 2}$, $e > 2 \rightarrow \ln e > \ln 2 \rightarrow 1 > \ln 2 \rightarrow \frac{1}{\ln 2} > 1$

$\rightarrow$ Series diverge.

Ex: Find all values of x where the series converge (radius of convergence) and then find sum

① $\sum_{k=0}^{\infty} 3(-1)^k \left(\frac{x}{2}\right)^k$ $\quad r = \frac{-x}{2}$, $a = 3$

$r = \frac{-x}{2}$ series conv. if $|r| < 1 \rightarrow \left|\frac{x}{2}\right| < 1 \rightarrow |x| < 2$

radius of conv. $(-2, 2)$ $\quad sum = \frac{a}{1-r} = \frac{3}{1+\frac{x}{2}} = \frac{6}{2+x}$

② $\sum_{k=0}^{\infty} \frac{2^k}{x^{k+2}}$, $\quad r = \frac{2}{x}$, $\quad a = \frac{1}{x^2}$

conv. if $|r| < 1 \rightarrow \left|\frac{2}{x}\right| < 1 \rightarrow |x| > 2 \rightarrow x > 2$ or $x < -2$

$sum = \frac{a}{1-r} = \frac{\frac{1}{x^2}}{1-\frac{2}{x}} = \frac{1}{x^2 - 2x}$

③ $\sum_{k=0}^{\infty} (x-3)^k$, $\quad a = 1$, $\quad r = x-3$

$|r| < 1 \rightarrow |x-3| < 1 \rightarrow -1 < x-3 < 1 \rightarrow 2 < x < 4$

$sum = \frac{a}{1-r} = \frac{1}{1-(x-3)} = \frac{1}{4-x} \rightarrow \sum_{k=0}^{\infty} (x-3)^k = \frac{1}{4-x}$

④ $\sum_{k=1}^{\infty} e^{-kx}$, $\quad a = e^{-x}$, $\quad r = e^{-x}$

$|r| < 1 \rightarrow |e^{-x}| < 1 \rightarrow e^{-x} < 1 \rightarrow e^x > 1 \rightarrow \ln e^x > \ln 1 \rightarrow x > 0$

$sum = \frac{a}{1-r} = \frac{e^{-x}}{1-e^{-x}} = \frac{1}{e^x - 1}$

3 Telescoping Series

$\sum_k u_k$

$S_n = u_1 + u_2 + \cdots + u_n$

find $\lim_{n\to\infty} S_n$, if $\lim_{n\to\infty} S_n = \infty \rightarrow$ Series diverge

if $\lim_{n\to\infty} S_n = S \rightarrow$ series conv. and $\sum_{k=c}^{\infty} u_k = S$

We can find $\lim S_n$ here by deleting some terms.

Examples:

① $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$

$= \sum_{k=1}^{\infty} \frac{1}{k} - \frac{1}{k+1}$

$S_n = u_1 + u_2 + \cdots + u_{n-1} + u_n$

$= 1 - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} + / \cdots / + \underbrace{\cancel{\frac{1}{n-1}} - \cancel{\frac{1}{n}}}_{u_{n-1}} + \underbrace{\cancel{\frac{1}{n}} - \frac{1}{n+1}}_{u_n}$

$u_1 \quad u_2 \quad u_3$

$S_n = 1 - \frac{1}{n+1} \rightarrow \lim_{n\to\infty} S_n = 1 = \text{sum} = \sum_{k=1}^{\infty} \frac{1}{k(k+1)}$

② show that $\frac{1}{1(3)} + \frac{1}{2(4)} + \frac{1}{3(5)} + \cdots = \frac{3}{4}$.

$\sum_{k=1}^{\infty} \frac{1}{k(k+2)} \underset{\text{Partial Fraction}}{=} \sum_{k=1}^{\infty} \frac{1/2}{k} - \frac{1/2}{k+2} = \frac{1}{2} \sum_{k=1}^{\infty} \frac{1}{k} - \frac{1}{k+2}$

$S_n = \frac{1}{2}\left[1 - \cancel{\frac{1}{3}} + \frac{1}{2} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{5}} + / \cdots / + \cancel{\frac{1}{n-1}} - \frac{1}{n+1} + \cancel{\frac{1}{n}} - \frac{1}{n+2}\right]$

$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{2}\left[1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right] = \frac{3}{4} = \text{Sum}$

(3) $\sum_{k=2}^{\infty} \ln\left(\frac{k}{k+1}\right) = \sum_{k=2}^{\infty} \ln k - \ln(k+1)$

$S_n = u_1 + u_2 + \cdots + u_n = \ln 2 - \ln 3 + \ln 3 - \ln 4 + \cdots + \ln(n-1) - \ln(n) + \ln(n) - \ln(n+1)$

$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \ln 2 - \ln(n+1) = -\infty$

→ Series diverge

(4) $\ln\left(1-\frac{1}{4}\right) + \ln\left(1-\frac{1}{9}\right) + \ln\left(1-\frac{1}{16}\right) + \cdots$

$= \sum_{k=2}^{\infty} \ln\left(1-\frac{1}{k^2}\right) = \sum \ln\left(\frac{k^2-1}{k^2}\right) = \sum \ln\left(\frac{(k-1)(k+1)}{k\,k}\right)$

$= \sum \ln\left(\frac{k-1}{k}\right) + \ln\left(\frac{k+1}{k}\right) = \sum_{k=2}^{\infty} \ln\left(\frac{k-1}{k}\right) - \ln\left(\frac{k}{k+1}\right)$

$S_n = \ln\frac{1}{2} - \ln\frac{2}{3} + \ln\frac{2}{3} - \ln\frac{3}{4} + \cdots + \ln\left(\frac{n-2}{n-1}\right) - \ln\left(\frac{n-1}{n}\right) + \ln\left(\frac{n-1}{n}\right) - \ln\left(\frac{n}{n+1}\right)$

$\lim_{n\to\infty} S_n = \lim \ln\frac{1}{2} - \ln\left(\frac{n}{n+1}\right)$

$= \ln\frac{1}{2} - \ln\left(\lim_{n\to\infty} \frac{n}{n+1}\right) = \ln\frac{1}{2} - \ln 1 = \ln\frac{1}{2}$

$= -\ln 2.$

$\therefore$ sum $= -\ln 2.$

Convergence tests

1) Divergence test

$\sum_{k=c}^{\infty} u_k$

If $\lim_{k\to\infty} u_k \neq 0$, then the series diverge

If $\lim_{k\to\infty} u_k = 0$, then test failed.

Examples:

① $\sum_{k=1}^{\infty} \frac{k}{k+1}$, $\lim_{k\to\infty} \frac{k}{k+1} = 1 \neq 0 \rightarrow$ series diverge

② $\sum_{k=2}^{\infty} \ln k$, $\lim_{k\to\infty} \ln k = +\infty \rightarrow$ diverge

③ $\sum_{k=3}^{\infty} \left(1+\frac{2}{k}\right)^k$, $\lim_{k\to\infty} \left(1+\frac{2}{k}\right)^k = e^2 \neq 0 \rightarrow$ diverge

④ $\sum \frac{(2k+2)^2}{1-3k^2}$, $\lim_{k\to\infty} \frac{(2k+2)^2}{1-3k^2} = \frac{-4}{3} \neq 0 \rightarrow$ diverge

⑤ $\sum k^2 \sin^2\left(\frac{1}{k}\right)$, $\lim_{k\to\infty} k^2 \sin^2\left(\frac{1}{k}\right) = \lim \frac{\sin^2\left(\frac{1}{k}\right)}{1/k^2} = 1 \rightarrow$ diverge

⑥ $\sum_{k=5}^{\infty} \frac{k}{\ln k}$, $\lim_{k\to\infty} \frac{k}{\ln k} = \lim_{k\to\infty} \frac{1}{1/k} = \ln k = \infty \rightarrow$ div.

⑦ $\sum_{k=2}^{\infty} \pi$, $\lim \pi = \pi \neq 0 \rightarrow$ div.

* if $\lim u_k = 0 \rightarrow$ test failed. We have to find another test. The series may conv. or div.

2 p-Series test

Series of the form

$$\sum \frac{1}{k^p} \quad , p>0$$

* If $0<p\leq 1$, then the series diverge
* If $p>1$, " " " converge

Examples:

① $\sum \frac{1}{k}$, $p=1 \rightarrow$ series diverge (This series is called harmonic series)

② $\sum \frac{1}{\sqrt{k}}$, $p=\frac{1}{2}<1 \rightarrow$ series div.

③ $\sum k^{-2} = \sum \frac{1}{k^2} \rightarrow p=2 \rightarrow$ converge.

④ $\sum \frac{1}{\sqrt{k^3}}$, $p=\frac{3}{2}>1 \rightarrow$ converge

⑤ $\sum \frac{k^2}{\sqrt{k^7}} = \sum \frac{1}{k^{\frac{7}{2}-2}} = \sum \frac{1}{k^{5/2}}$, $p=\frac{5}{2}>1 \rightarrow$ Conv.

⑥ $\sum \frac{1}{k^{\pi}}$, $p=\pi>1 \rightarrow$ Converge.

Theorem:

(a) If $\sum u_k$ and $\sum v_k$ both converge, then $\sum (u_k \pm v_k)$ is convergent and

$$\sum (u_k \pm v_k) = \sum u_k \pm \sum v_k$$

(sum $\sum u_k \pm v_k$ = sum $\sum u_k \pm$ sum $\sum v_k$)

Ex: find $\sum_{k=1}^{\infty} \left[\left(\frac{1}{2}\right)^k - \frac{1}{k(k+1)} \right] = \underbrace{\sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k}_{\text{geometric}} + \underbrace{\sum \frac{1}{k(k+1)}}_{\text{telescoping}}$

$= \frac{1}{2} + 1 = \frac{3}{2}$.

(b) If $\sum u_k$ conv., then $\sum c u_k$ conv.

and $\sum cu_k = c \sum u_k$

If $\sum u_k$ div., then $\sum c u_k$ div.

(c) $\sum_{k=1}^{\infty} u_k$ and $\sum_{k=c}^{\infty} u_k$ both conv. or both diverge

if converge they do not have the same sum.

(d) If $\sum u_k$ converge, then $\lim_{k \to \infty} u_k = 0$

3 Integral test

Let $\sum u_k$ be series of +ve terms,

$f(k) = u_k$ is decreasing and cont. on some interval $[a, \infty)$. Then

$\int_a^\infty f(x)\,dx$ and $\sum u_k$ both conv. or both div.

Examples:

(1) $\sum_{k=1}^{\infty} \frac{1}{5k+2}$, $f(x) = \frac{1}{5x+2}$ cont. on $[1,\infty)$

$f' = \frac{-5}{(5x+2)^2} < 0 \rightarrow f$ decreasing, $\int_1^\infty \frac{dx}{5x+2} = \frac{1}{5}\ln|5x+2|\Big|_1^\infty = \infty$

integral diverge $\Rightarrow$ series diverge

(2) $\sum_{k=1}^{\infty} \frac{1}{1+9k^2}$, $f(x) = \frac{1}{1+9x^2}$ cont. $f' = \frac{-18x}{(1+9x)^2} < 0$ on $[1,\infty)$

$\int_1^\infty \frac{dx}{1+9x^2} \underset{\substack{z=3x \\ dz=3dx}}{=} \frac{1}{3}\int \frac{dz}{1+z^2} = \frac{1}{3}\tan^{-1}3x\Big|_1^\infty = \frac{1}{3}\left(\frac{\pi}{2} - \tan^{-1}3\right) \neq \infty \rightarrow$ conv.

$\rightarrow$ series conv.

(3) $\sum_{k=3}^{\infty} \frac{\ln k}{k}$, $f(x) = \frac{\ln x}{x}$ cont. on $[3,\infty)$

$f' = \frac{1-\ln x}{x^2} < 0$ on $[3,\infty)$

$\int_3^\infty \frac{\ln x}{x}\,dx = \frac{(\ln x)^2}{2}\Big|_3^\infty = \infty \rightarrow$ integral diverge

$\rightarrow$ series diverge

(4) $\sum_{k=1}^{\infty} \frac{\tan^{-1} k}{1+k^2}$, $f(x) = \frac{\tan^{-1} x}{1+x^2}$ cont.

$f' = \frac{1 - 2x\tan^{-1} x}{(1+x^2)^2} < 0$ on $[1, \infty)$

$x \geq 1 \to \tan^{-1} x \geq \tan^{-1} 1 = \frac{\pi}{4}$

$\to x\tan^{-1} x \geq \frac{\pi}{4}$

$-2x\tan^{-1} x \leq \frac{\pi}{2}$

$1 - 2x\tan^{-1} x \leq 1 - \frac{\pi}{2}$ ($\sim < 0$)

$$\int_1^{\infty} \frac{\tan^{-1} x}{1+x^2}\,dx = \frac{(\tan^{-1} x)^2}{2}\Big|_1^{\infty} = \frac{1}{2}\left[\left(\frac{\pi}{2}\right)^2 - \left(\frac{\pi}{4}\right)^2\right] \neq \infty$$

$\to$ integral converge $\to$ series converge

(5) $\sum_{k=1}^{\infty} \frac{k}{1+k^2}$

$f(x) = \frac{x}{1+x^2}$ cont. on $[1, \infty)$

$1 - x^2$: $-$ | $+$ | $-$ (at -1, 1)

$f' = \frac{1 - x^2}{(1+x^2)^2} < 0$ on $[1, \infty)$

$$\int_1^{\infty} \frac{x}{1+x^2}\,dx = \frac{1}{2}\ln(1+x^2)\Big|_1^{\infty} = \infty$$

$\to$ integral diverge

$\to$ series diverge.

4 Comparison test

let $\sum a_k$, $\sum b_k$ be two series of nonnegative terms and $a_k \le b_k$, $k \ge 1$. Then

a) If $\sum b_k$ (the bigger) conv., then $\sum a_k$ (the smaller) conv.

b) If $\sum a_k$ (the smaller) div, then $\sum b_k$ (the bigger) div.

Examples:

① $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}-2}$, take $\sum \frac{1}{\sqrt{k}}$ $p=\frac{1}{2}$ div.

$\sqrt{k}-2 < \sqrt{k} \rightarrow \frac{1}{\sqrt{k}-2} > \frac{1}{\sqrt{k}}$ smaller div. → bigger div.

② $\sum_{k=1}^{\infty} \frac{1}{2k^2+k}$, take $\sum \frac{1}{k^2}$ $p=2$ conv.

$2k^2+k > 2k^2 \rightarrow \frac{1}{2k^2+k} < \frac{1}{2k^2}$ bigger conv. → smaller conv.

③ $\sum_{k=2} \frac{1}{3^k+5}$, take $\sum \frac{1}{3^k}$ conv. geometric

$3^k+5 > 3^k \rightarrow \frac{1}{3^k+5} < \frac{1}{3^k}$ → $\sum \frac{1}{3^k}$ conv. (bigger)

→ $\sum \frac{1}{3^k+5}$ conv.

④ $\sum \frac{4+|\cos k|}{k^3}$, $|\cos k| < 1 \rightarrow \frac{4+|\cos k|}{k^3} < \frac{5}{k^3}$

$\sum \frac{5}{k^3}$ conv. p-series

bigger conv. → smaller $\sum \frac{4+|\cos k|}{k^3}$ conv.

5 Limit comparison test

Let $\sum a_k$, $\sum b_k$ be series of +ve terms

let $r = \lim_{k\to\infty} \frac{a_k}{b_k}$

(1) If $0 < r < \infty$, then both conv. or both div.

(2) If $r = 0$ and $\sum b_k$ conv., then $\sum a_k$ conv.

(3) If $r = \infty$ and $\sum b_k$ div., then $\sum a_k$ div.

Examples: Test conv. or diverge

① $\sum \frac{1}{\sqrt{k}+1}$, take $b_k = \frac{1}{\sqrt{k}}$, $\sum \frac{1}{\sqrt{k}}$ div p-series

$$r = \lim_{k\to\infty} \frac{a_k}{b_k} = \lim \frac{\frac{1}{\sqrt{k}+1}}{\frac{1}{\sqrt{k}}} = \lim_{k\to\infty} \frac{\sqrt{k}}{\sqrt{k}+1} = 1 \to \sum \frac{1}{\sqrt{k}+1} \text{ div.}$$

② $\sum \frac{1}{2k^2+k}$, take $b_k = \frac{1}{k^2}$ conv. p-series

$$r = \lim \frac{a_k}{b_k} = \lim \frac{k^2}{2k^2+k} = \frac{1}{2} \to \sum \frac{1}{2k^2+k} \text{ conv.}$$

③ $\sum \frac{3k^3-2k^2+4}{k^7-k^3+2}$, take $b_k = \frac{1}{k^4}$, $\sum \frac{1}{k^4}$ conv. p-series

$$r = \lim \frac{a_k}{b_k} = 3 \to \sum \frac{3k^3-2k^2+4}{k^7-k^3+2} \text{ conv.}$$

④ $\sum \sin\left(\frac{\pi}{k}\right)$, take $b_k = \frac{\pi}{k}$ $\sum \frac{\pi}{k}$ div. p-series

$$r = \lim_{k\to\infty} \frac{\sin(\frac{\pi}{k})}{\pi/k} = \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1 \to \sum \sin\left(\frac{1}{k}\right) \text{ div.}$$

6 Ratio test

Let $\sum u_k$ be series of +ve terms

let $r = \lim_{k\to\infty} \frac{u_{k+1}}{u_k}$

(a) If $0 \le r < 1$, then series converge.

(b) If $r > 1$ or ∞, then series diverge

(c) If $r = 1$, test failed.

Examples:

(1) $\sum \frac{1}{k!}$, $r = \lim \frac{\frac{1}{(k+1)!}}{\frac{1}{k!}} = \lim \frac{k!}{(k+1)k!} = 0$

→ series converge.

(2) $\sum \frac{k}{2^k}$, $r = \lim \frac{\frac{k+1}{2^{k+1}}}{\frac{k}{2^k}} = \lim \frac{(k+1)2^k}{k\,2^{k+1}} = \lim \frac{k+1}{2k} = \frac{1}{2}$

→ series converge.

(3) $\sum \frac{k^k}{k!}$, $r = \lim \frac{(k+1)^{k+1}}{(k+1)!} \cdot \frac{k!}{k^k} = \lim \frac{(k+1)(k+1)^k}{(k+1)k!} \frac{k!}{k^k} = \left(\frac{k+1}{k}\right)^k$

$= \lim_{k\to\infty} \left(1+\frac{1}{k}\right)^k = e^1 > 1$ → series diverge.

(4) $\sum \frac{\ln k}{e^k}$, $r = \lim_{k\to\infty} \frac{\ln(k+1)}{e^{k+1}} \cdot \frac{e^k}{\ln k} = \frac{1}{e} \lim \frac{\ln(k+1)}{\ln k} = \frac{1}{e} < 1$ conv.

(5) $\sum \frac{(k!)^2}{(2k)!}$, $r = \lim \frac{((k+1)!)^2}{(2(k+1))!} \cdot \frac{(2k)!}{(k!)^2}$

$= \lim \frac{(k+1)k!(k+1)k!}{(2k+2)(2k+1)(2k)!} \cdot \frac{(2k)!}{k!k!} = \lim \frac{(k+1)(k+1)}{(2k+2)(2k+1)} = \frac{1}{4} < 1$

→ conv.

7 Root test

let $\sum u_k$ be series of +ve terms

$$r = \lim_{k\to\infty} (u_k)^{\frac{1}{k}}$$

(a) If $0 \le r < 1$, then series conv.

(b) If $r > 1$ or $+\infty$, then series diverge.

(c) If $r = 1$, then test failed.

Examples.

(1) $\sum_{k=2}^{\infty} \left(\frac{4k-5}{2k+1}\right)^k$, $r = \lim \frac{4k-5}{2k+1} = 2 > 1 \to$ diverge.

(2) $\sum \frac{1}{(\ln(k+1))^k}$, $r = \lim \frac{1}{\ln(k+1)} = 0 \to$ Converge

(3) $\sum \left(\frac{k}{k+1}\right)^{k^2}$, $r = \lim \left(\frac{k}{k+1}\right)^k = \lim \frac{1}{\left(1+\frac{1}{k}\right)^k} = \frac{1}{e} < 1$
$\to$ converge

(4) $\sum$ [illegible], $r = \lim$ [illegible] $=$

Alternating Series

Def:

An alternating series are series that have +ve and -ve terms.

like $\sum (-1)^k a_k$, $\sum \cos \pi k \, a_k$, $\sum \sin\left(\frac{(2k-1)\pi}{2}\right) a_k$

* Alternating series test

Let $\sum (-1)^k a_k$ is an alternating series and

(a) $a_1 \geq a_2 \geq a_3 \geq \cdots$ $\left(a_k \geq a_{k+1}, k \geq 1\right)$ $\left(\frac{a_{k+1}}{a_k} \leq 1\right)$

(b) $\lim_{k \to \infty} a_k = 0$

then the alternating series is converge

Examples:

① $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k}$, $k+1 > k \rightarrow \frac{1}{k} < \frac{1}{k+1}$ and $\lim \frac{1}{k} = 0$

$\Rightarrow$ series converge.

② $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k+3}{k(k+1)}$, $\frac{a_{k+1}}{a_k} = \frac{k+4}{(k+1)(k+2)} \cdot \frac{k(k+1)}{k+3}$

$= \frac{k^2+4k}{k^2+4k+k+6} < 1$

$\lim_{k \to \infty} \frac{k+3}{k(k+1)} = 0 \rightarrow$ series converge

(3) $\sum (-1)^k \frac{k}{3^k}$

$$\frac{a_{k+1}}{a_k} = \frac{k+1}{3^{k+1}} \cdot \frac{3^k}{k} = \frac{k+1}{3k} < \frac{k+k}{3k} = \frac{2}{3} < 1$$

$$\lim_{k\to\infty} \frac{k}{3^k} = \lim_{k\to\infty} \frac{1}{3^k \ln 3} = 0 \rightarrow \text{Series converge}$$

(4) $\sum_{k=1}^{\infty} (\cos k\pi)\, e^{-k}$,

is alternating series $= \sum (-1)^k e^{-k}$

$$\frac{a_{k+1}}{a_k} = \frac{e^{-(k+1)}}{e^{-k}} = e^{-1} = \frac{1}{e} < 1$$

$$\lim_{k\to\infty} e^{-k} = 0 \rightarrow \text{Series converge.}$$

Absolute Convergence

Def.

$\sum u_k$ is converge absolutely if $\sum |u_k|$ converge

$\sum u_k$ is diverge absolutely if $\sum |u_k|$ diverge

Examples:

(1) $\sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{2^k}$

$\sum |u_k| = \sum \frac{1}{2^k}$ geometric series $r = \frac{1}{2}$ → converge

→ $\sum \frac{(-1)^{k+1}}{2^k}$ converge absolutely.

(2) $\sum_{k=3}^{\infty} \frac{(-1)^{k+1}}{k}$

$\sum \frac{1}{k}$ diverge p-series ⇒ $\sum \frac{(-1)^{k+1}}{k}$ diverge absolutely

Theorem:

If $\sum |u_k|$ converge then $\sum u_k$ converge.

(that is conv. absolutely → converge).

So Ex1: $\sum \frac{(-1)^{k+1}}{2^k}$ converge.

div absolutely ↛ div.

conv. ↛ conv. abs.

Ex. $\sum_{k=1}^{\infty} \frac{\cos k}{k^2}$

$\left|\frac{\cos k}{k^2}\right| \leq \frac{1}{k^2}$ conv. $\Rightarrow \sum \left|\frac{\cos k}{k^2}\right|$ conv.

$\Rightarrow \sum \frac{\cos k}{k^2}$ conv. absolutely

$\Rightarrow \sum \frac{\cos k}{k^2}$ converge.

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Conditionally Convergence

If $\sum u_k$ converge and diverge absolutely, then we say that $\sum u_k$ converge conditionally.

Examples:

① $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$

$\sum \left|\frac{(-1)^{k+1}}{k}\right| = \sum \frac{1}{k}$ diverge $\rightarrow \sum \frac{(-1)^{k+1}}{k}$ div. absolutely

and

$\sum \frac{(-1)^{k+1}}{k}$ alternating series $\frac{1}{k} > \frac{1}{k+1}$, $\lim_{k\to\infty} \frac{1}{k} = 0$

so $\sum \frac{(-1)^{k+1}}{k}$ converge $\Rightarrow \sum \frac{(-1)^{k+1}}{k}$ converge conditionally

② $\sum (-1)^{k+1} \frac{k+3}{k(k+1)}$

$\sum a_k = \sum \frac{k+3}{k(k+1)}$, $\sum b_k = \sum \frac{1}{k}$ div., $r = \lim \frac{a_k}{b_k} = 1 < \infty \rightarrow$ both div.

$\Rightarrow \sum \frac{k+3}{k(k+1)}$ diverge.

$\sum (-1)^k \frac{k+3}{k(k+1)}$ alternating series $a_k < a_{k+1}$, $\lim a_k = 0 \rightarrow$ conv.

$\Rightarrow \sum (-1)^{k+1} \frac{k+3}{k(k+1)}$ conv. conditionally.

③ $\sum_{k=3}^{\infty} (-1)^k \frac{\ln k}{k}$, $\ln k > 1$ $k > 3 > e \rightarrow \frac{\ln k}{k} > \frac{1}{k}$

smaller div. $\rightarrow \sum \frac{\ln k}{k}$ div.

$\sum (-1)^k \frac{\ln k}{k}$ altern. series conv.

$\Rightarrow \sum (-1)^k \frac{\ln k}{k}$ conv. conditionally.

Ratio test for absolute Convergence

Let $\sum u_k$ be series of nonzero terms.

$$r = \lim_{k\to\infty} \frac{|u_{k+1}|}{|u_k|}$$

(a) If $r<1$, then $\sum u_k$ conv. absolutely $\to$ conv.

(b) if $r>1$ or ∞, then $\sum u_k$ diverge

(c) if $r=1$ test failed.

Examples:

(1) $\sum_{k=1}^{\infty} \frac{(-1)^k 2^k}{k!}$, $r = \lim_{k\to\infty} \frac{|u_{k+1}|}{|u_k|} = \lim_{k\to\infty} \frac{2}{k+1} = 0 < 1$

$\to$ series conv. abs. $\to$ conv.

(2) $\sum_{k=1}^{\infty} (-1)^k \frac{(2k-1)!}{3^k}$

$r = \lim_{k\to\infty} \frac{|u_{k+1}|}{|u_k|} = \frac{1}{3}\lim_{k\to\infty} (2k+1)(2k) = \infty \to$ series diverge

(3) $\sum \frac{(-1)^k \cdot k}{5^k}$

$r = \frac{1}{5} < 1 \to$ conv. absolutely $\to$ conv.

Examples:

① $\sum \frac{5\sin^2 k}{k!}$

$\frac{5\sin^2 k}{k!} < \frac{5}{k!}$, $\sum \frac{5}{k!}$ conv. ratio test

bigger conv. → smaller $\sum \frac{5\sin^2 k}{k!}$ conv.

② $\sum \frac{\ln k}{k}$

$\ln k > 1$ for $k \geq 3$ → $\frac{\ln k}{k} > \frac{1}{k}$,

$\sum \frac{1}{k}$ div. Smaller div. → $\sum \frac{\ln k}{k}$ div.

③ $\sum \frac{4}{2+3^k k}$

$\frac{4}{2+3^k k} < \frac{4}{3^k k}$, $\sum \frac{4}{3^k k}$ conv. ratio test

→ $\sum \frac{4}{2+3^k k}$ conv.

④ $\sum_{k=1}^{\infty} \frac{(-1)^k}{5k^2-k}$

$\frac{1}{5k^2-k} < \frac{1}{5k^2-k^2} = \frac{1}{4k^2}$, $\sum \frac{1}{4k^2}$ conv. (bigger)

→ $\sum \frac{1}{5k^2-k}$ conv. → $\sum \frac{(-1)^k}{5k^2-k}$ abs. conv. → conv.

Maclaurian and Taylor Series

Def. (Maclaurian Series)

If f has all its derivatives at $x=0$, then the Maclaurian series for $f(x)$ is

$$\sum_{k=0}^{\infty} f^{(k)}(0)\frac{x^k}{k!} = f(0) + f'(0)\frac{x}{1!} + f''(0)\frac{x^2}{2!} + \cdots$$

Def. (Taylor Series)

If f has all its derivatives at $x=c$, then the Taylor series for $f(x)$ at $x=c$ is

$$\sum_{k=0}^{\infty} f^{(k)}(c)\frac{(x-c)^k}{k!} = f(c) + f'(c)(x-c) + f''(c)\frac{(x-c)^2}{2!} + \cdots$$

Examples:

Maclaurian Series for

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \quad \sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}, \quad \cosh x = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$$

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

Examples: Find Maclaurin series for

① $e^{3x} = \sum_{k=0}^{\infty} \frac{(3x)^k}{k!} = \sum_{k=0}^{\infty} \frac{3^k x^k}{k!}$

② $\sin \pi x = \sum_{k=0}^{\infty} (-1)^k \frac{\pi^{2k+1} x^{2k+1}}{(2k+1)!}$

③ $\cos\left(\frac{\pi}{2}x\right) = \sum (-1)^k \left(\frac{\pi}{2}\right)^{2k} \frac{x^{2k}}{(2k)!} = \sum (-1)^k \frac{\pi^{2k} x^{2k}}{4^k (2k)!}$

④ $xe^x = \sum_{k=0}^{\infty} \frac{x \cdot x^k}{k!} = \sum_{k=0}^{\infty} \frac{x^{k+1}}{k!} = \sum_{k=1}^{\infty} \frac{x^k}{(k-1)!}$

⑤ $\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} = \sum_{k=0}^{\infty} (-1)^k \frac{x^k}{k!}$

Ex: Find Taylor series for ⓐ $\frac{1}{x}$ at $x=1$

$$= \sum \frac{f^{(k)}(1)(x-1)^k}{k!}$$

$f(1)=1$, $f'(1)=-1$, $f''(1)=2$, $f'''(1)=-6$, $f^{(4)}(1)=24$

$\cdots \Rightarrow f^{(k)}(1) = (-1)^k k!$ $\quad k \geq 0$.

$$\Rightarrow \frac{1}{x} = \sum \frac{(-1)^k k! (x-1)^k}{k!} = \sum_{k=0}^{\infty} (-1)^k (x-1)^k$$

ⓑ $f(x) = \ln x$, $x = e$

$f(e)=1$, $f'(e)=e^{-1}$, $f''(e)=-e^{-2}$, $f'''(e)=2e^{-3}$, $f^{(4)}(e)=-6e^{-4}$,

$\cdots \rightarrow f^{(k)}(e) = (-1)^{k-1}(k-1)!\, e^{-k}$, $k \geq 1$

$$\ln x = f(e) + \sum \frac{f^{(k)}(e)(x-e)^k}{k!} = 1 + \sum \frac{(-1)^{k-1}(k-1)! e^{-k} (x-e)^k}{k!}$$

$$= 1 + \sum_{k=1}^{\infty} (-1)^{k-1} \frac{(x-e)^k}{ek}$$

Examples: Find radius of convergence

① $\sum_{k=0}^{\infty} 3^k x^k$ by geometric or ratio test

$r = 3x$ if $|3x| < 1$ → series conv.

$|x| < \frac{1}{3}$ → radius of conv. $= \frac{1}{3}$.

② $\sum_{k=0}^{\infty} \frac{x^k}{k+1}$, $r = \lim \frac{u_{k+1}}{u_k} = \lim \frac{k+1}{k+2}|x| < 1$

radius of conv. $= 1$, interval of conv. $(-1, 1)$

③ $\sum \frac{(-1)^k x^{2k}}{(2k)!}$, $r = \lim \frac{|x|^2}{(2k+2)(2k+1)} = 0$ conv. for all x

radius of conv is ∞ interval of conv. $(-\infty, \infty)$

④ $\sum_{k=0}^{\infty} \frac{(x-3)^k}{2^k}$

$r = \left|\frac{x-3}{2}\right| < 1$ → $1 < x < 5$

→ interval of conv. $(+1, 5)$

Summary of Convergence tests

① Divergence test

If $\lim_{k\to\infty} u_k \neq 0 \Rightarrow$ series diverge.

② Integral test

$\sum u_k$ +ve terms, $f(k) = u_k$ decreas cont. on $[a, \infty)$, then

$\sum u_k$ and $\int f(x)\,dx$ both conv. or both div.

③ Comparison test

$\sum a_k$, $\sum b_k$ series of nonnegative terms

$a_k \leq b_k$, $k \geq 1$ $\Rightarrow$ If $\sum b_k$ conv. $\rightarrow$ $\sum a_k$ conv.

If $\sum a_k$ div $\rightarrow$ $\sum b_k$ div.

④ Limit Comparison test

$\sum a_k$, $\sum b_k$ +ve terms, $r = \lim_{k\to\infty} \frac{a_k}{b_k}$

If $0 < r < \infty$, then both conv. or both div.

⑤ Ratio test

$\sum u_k$ +ve terms, $r = \lim_{k\to\infty} \frac{u_{k+1}}{u_k}$, then

a) if $r < 1 \rightarrow$ series conv.
b) if $r > 1$ or $\infty \rightarrow$ div.
c) if $r = 1$ test failed

⑥ Root test

$\sum u_k$ +ve terms, $r = \lim_{k\to\infty} (u_k)^{\frac{1}{k}}$

a) if $r < 1 \rightarrow$ conv.
b) if $r > 1$ or $\infty \rightarrow$ div.
c) if $r = 1$ test failed

⑦ Alternating Series test

$\sum (-1)^k a_k$ if a) $a_k \geq a_{k+1}$, $k \geq 1$, then the series conv.

b) $\lim_{k\to\infty} a_k = 0$

⑧ Ratio test for absolutely conv.

* If $\sum a_k = \sum |(-1)^k a_k|$ conv. $\rightarrow$ $\sum (-1)^k a_k$ abs. conv. and so conv.
* If $\sum a_k = \sum |(-1)^k a_k|$ div. $\rightarrow$ $\sum (-1)^k a_k$ abs. diverge.
* If $\sum (-1)^k a_k$ conv. and $\sum a_k$ div. $\rightarrow$ $\sum (-1)^k a_k$ conditionally conv.

$\sum u_k$ +ve and -ve terms

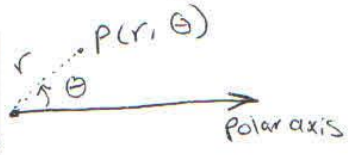
$r = \lim_{k\to\infty} \frac{|u_{k+1}|}{|u_k|}$

a) if $r < 1 \rightarrow$ series abs. conv. $\rightarrow$ conv.
b) if $r > 1$ or $\infty \rightarrow$ series div.
c) if $r = 1 \rightarrow$ test failed

Polar Coordinates

Def:

Polar coordinate is plane consist of fixed polar Point O called the pole (or origin) and a ray called polar axis



Any point in polar coordinate has the coordinate (r, θ)

r: distance from O to P (radial coordinate)

θ: angle from polar axis to $\overline{OP}$ (angular coordinate)
(counter clockwise if $\theta > 0$)

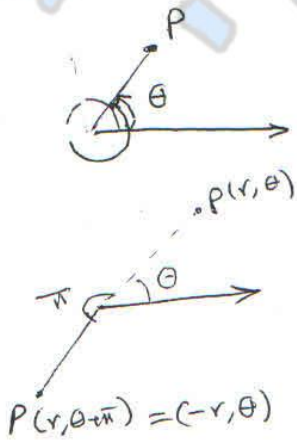
pole $O = (0, \theta)$ ← origin

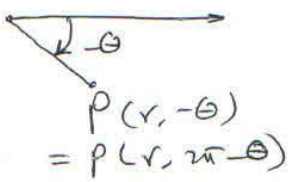
Note that

$$(r, \theta) = (r, \theta \pm 2n\pi)$$

$$(-r, \theta) = (r, \theta + \pi)$$

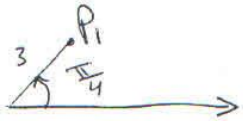
$$(r, -\theta) = (r, 2\pi - \theta)$$



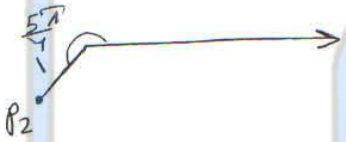


Ex: Plot the points in polar coordinate.

(a) $P_1\left(3, \frac{\pi}{4}\right)$



(b) $P_2\left(1, \frac{5\pi}{4}\right)$

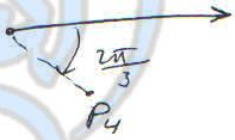


(c) $P_3(2, \pi)$

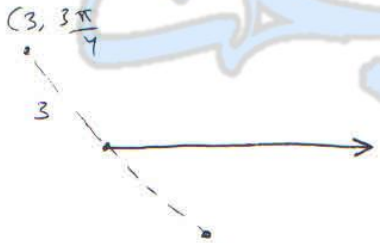


(d) $P_4\left(2, -\frac{2\pi}{3}\right)$

$= \left(2, 2\pi - \frac{2\pi}{3}\right) = \left(2, \frac{4\pi}{3}\right)$



(e) $P_5\left(-3, \frac{3\pi}{4}\right)$



$P_5\left(-3, \frac{3\pi}{4}\right)$

$= \left(3, \frac{3\pi}{4} + \pi\right)$

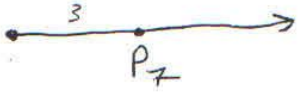
$= \left(3, \frac{7\pi}{4}\right)$

(f) $P_6(5, 3\pi) = (5, 3\pi - 2\pi)$

$= (5, \pi)$

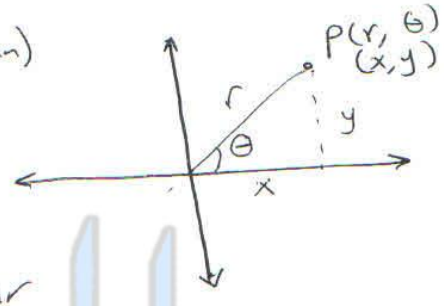


(g) $P_7(3, 0)$



Relationship between polar and rectangular Coordinates

From polar to rectangular (cartesian)



$$x = r\cos\theta$$
$$y = r\sin\theta$$

From Cartesian (rectangular) to polar

$$r^2 = x^2 + y^2$$
$$\tan\theta = \frac{y}{x} \rightarrow \theta = \tan^{-1}\frac{y}{x}.$$

Ex: Convert from polar to cartesian

① $(6, \frac{2\pi}{3})$ $x = 6\cos\frac{2\pi}{3} = 6(-\frac{1}{2}) = -3$, $y = 6\sin\frac{2\pi}{3} = 6(\frac{\sqrt{3}}{2}) = 3\sqrt{3}$ → $(x, y) = (-3, 3\sqrt{3})$

② $(3, 3\pi)$ $x = 3\cos 3\pi = -3$, $y = 3\sin 3\pi = 0$ → $(-3, 0)$

③ $(2, -\frac{2\pi}{3})$ $x = 2\cos(-\frac{2\pi}{3}) = -1$, $y = 2\sin(-\frac{2\pi}{3}) = 2 \cdot -\frac{\sqrt{3}}{2} = -\sqrt{3}$ → $(-1, -\sqrt{3})$

④ $(-\sqrt{2}, \frac{5\pi}{4})$ $x = -\sqrt{2}(\frac{-1}{\sqrt{2}}) = 1$, $y = -\sqrt{2}(\frac{-1}{\sqrt{2}}) = 1$ → $(1, 1)$

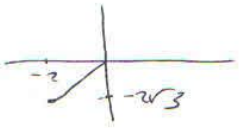
⑤ $(-2, \frac{\pi}{4})$ $x = -2\frac{1}{\sqrt{2}} = -\sqrt{2}$, $y = -2\frac{1}{\sqrt{2}} = -\sqrt{2}$ → $(-\sqrt{2}, -\sqrt{2})$

⑥ $(-5, 0)$ → $x = -5(1) = -5$, $y = -5(0) = 0$ → $(-5, 0)$

⑦ $(-4, -\frac{3\pi}{2})$ → $x = 0$, $y = -4$ → $(0, -4)$

Ex: Convert from Cartesian to polar

① $(-2, -2\sqrt{3})$



$r^2 = 16 \rightarrow r = 4$ $\quad \tan\theta = \frac{y}{x} = \sqrt{3}$

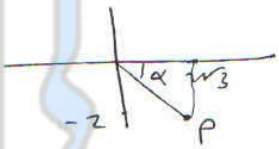
$\rightarrow \theta = \frac{\pi}{3}$ in 3rd quadrant

$\rightarrow \theta = \pi + \frac{\pi}{3} \rightarrow (r, \theta) = (4, \frac{4\pi}{3})$

$= (-4, \frac{\pi}{3})$

$(4, \frac{4\pi}{3} \pm 2n\pi)$ or $(-4, \frac{\pi}{3} \pm 2n\pi)$.

② $(2\sqrt{3}, -2)$



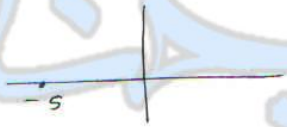
$r^2 = 12 + 4 = 16 \rightarrow r = 4$

$\tan\theta = \frac{-1}{\sqrt{3}} \rightarrow \alpha = \frac{\pi}{6} \rightarrow \theta = 2\pi - \frac{\pi}{6} = 11\frac{\pi}{6}$ $\quad \alpha = \frac{\pi}{6}$

$(4, 11\frac{\pi}{6} \pm 2n\pi)$.

$(-4, \frac{5\pi}{6} \pm 2n\pi)$.

③ $(-5, 0)$

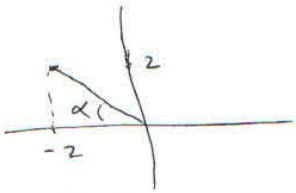


$r = 5 \quad \theta = \pi$

$(5, \pi)$ or $(-5, 0)$

$(5, \pi \pm 2n\pi)$ or $(-5, \pm 2n\pi)$.

④ $(-2, 2)$



$r^2 = 4 + 4 = 8 \rightarrow r = 2\sqrt{2}$

$\tan\alpha = 1 \rightarrow \alpha = \frac{\pi}{4}$

$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

$(2\sqrt{2}, \frac{3\pi}{4})$ or $(-2\sqrt{2}, \frac{7\pi}{4})$

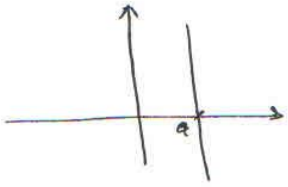
$(2\sqrt{2}, \frac{3\pi}{4} \pm 2n\pi)$ or $(-2\sqrt{2}, \frac{7\pi}{4} \pm 2n\pi)$.

Graphs in Polar coordinates

* Lines

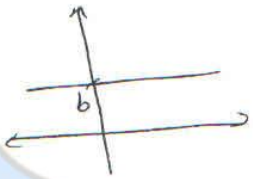
1) vertical line

$x = a \iff r\cos\theta = a$ or $r = a\sec\theta$



2) Horizontal line

$y = b \iff r\sin\theta = b$ or $r = b\csc\theta$



3) Linear equations

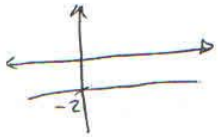
$\theta = \theta_0 \iff \tan\theta = \tan\theta_0 = c \iff \frac{y}{x} = c \Rightarrow y = cx$

Examples: sketch

a) $r\cos\theta = 3 \Rightarrow x = 3$



b) $r = -2\csc\theta \Rightarrow r\sin\theta = -2 \rightarrow y = -2$



c) $\theta = \frac{2\pi}{3} \Rightarrow \tan\theta = \tan\frac{2\pi}{3} \rightarrow \frac{y}{x} = \frac{-1}{\sqrt{3}}$

$\Rightarrow y = \frac{-1}{\sqrt{3}}x$



4) Circles

* $r = a \Rightarrow r^2 = a^2 \Rightarrow x^2 + y^2 = a^2$

circle Center $(0,0)$
radius $= a$

or * $r = 2a\cos\theta \Rightarrow r^2 = 2ar\cos\theta \Rightarrow x^2 + y^2 = 2ax$

$x^2 - 2ax + y^2 = 0 \rightarrow x^2 - 2ax + a^2 + y^2 = a^2 \rightarrow (x-a)^2 + y^2 = a^2$

circle Center $(a,0)$ radius a.

if $r = -2a\cos\theta \rightarrow$ circle Center $(-a,0)$ $r = a$

or * $r = 2a\sin\theta \Rightarrow r^2 = 2ar\sin\theta \Rightarrow x^2 + y^2 = 2ay$

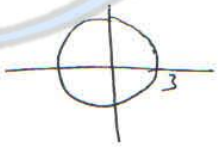
$x^2 + y^2 - 2ay + a^2 = a^2 \rightarrow x^2 + (y-a)^2 = a^2$

circle Center $(0,a)$ radius a.

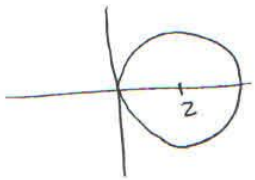
if $r = -2a\sin\theta \rightarrow$ circle Center $(0,-a)$ $r = a$

Examples

① $r = 3 \Rightarrow x^2 + y^2 = 9$

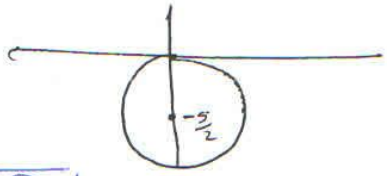


② $r = 4\cos\theta$ $\quad 2a = 4 \rightarrow a = 2 \Rightarrow$ circle Center $(2,0)$ $r = 2$



③ $r = -5\sin\theta$ $\quad$ circle Center $(0, -\frac{5}{2})$ $\quad r = \frac{5}{2}$

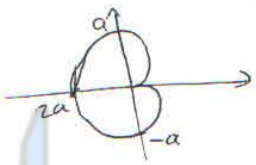
$2a = 5$
$a = \frac{5}{2}$



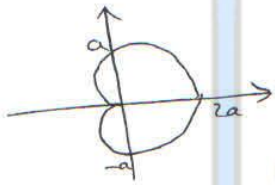
* Cardiods and Limacons

Cardiods

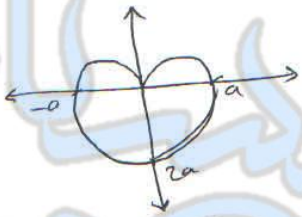
$r = a - a\cos\theta \quad , a>0$



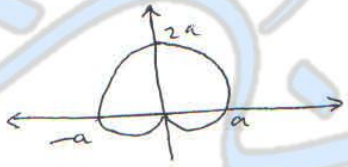
$r = -a + a\cos\theta \quad , a>0$



$r = a - a\sin\theta \quad , a>0$

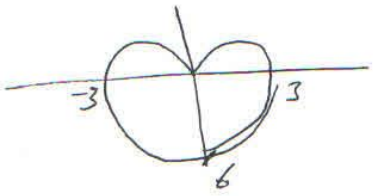


$r = -a + a\sin\theta \quad , a>0$

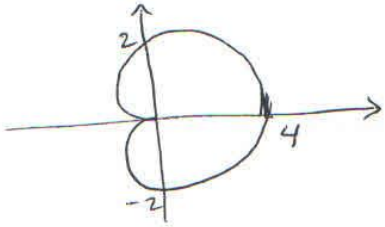


Ex: Sketch

a) $r = 3 - 3\sin\theta$



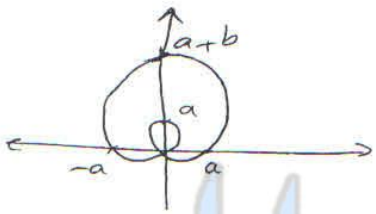
b) $r = -2 + 2\cos\theta$



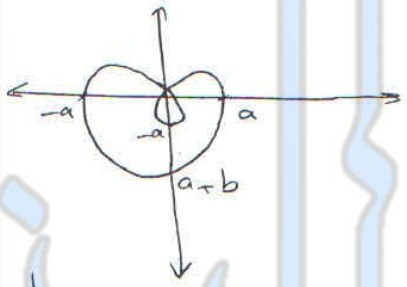
Limacons

$r = a \pm b\sin\theta$
$r = a \pm b\cos\theta$
} $a < b$

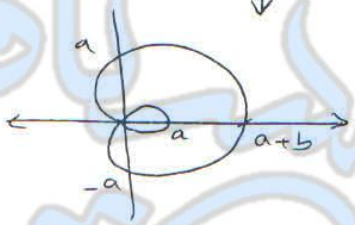
$r = a + b\sin\theta$ →



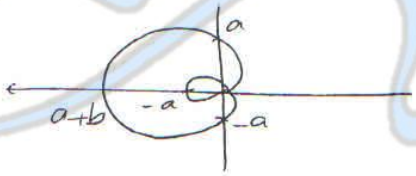
$r = a - b\sin\theta$ →



$r = a + b\cos\theta$

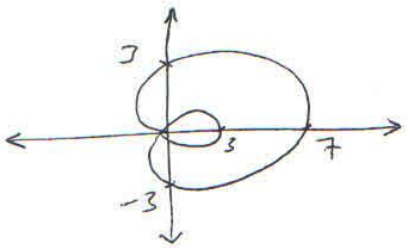


$r = a - b\cos\theta$

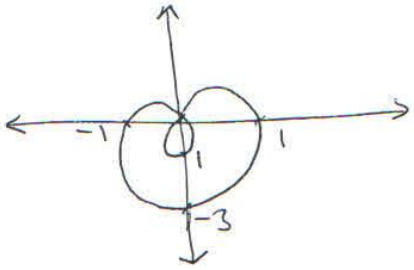


Ex: Sketch

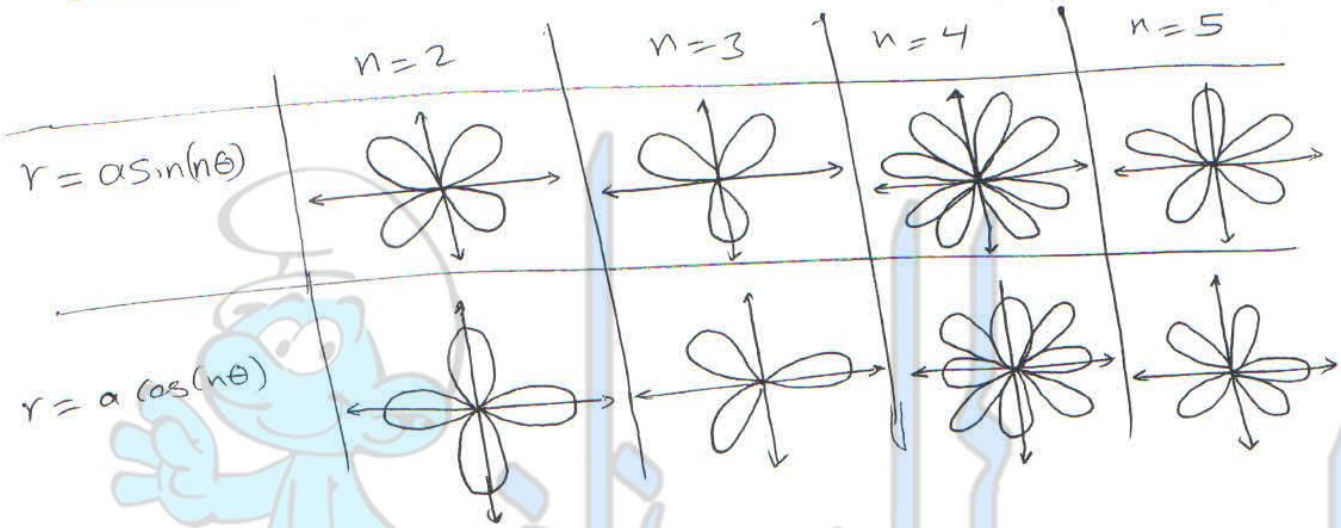
① $r = 3 + 4\cos\theta$



② $r = 1 - 2\sin\theta$



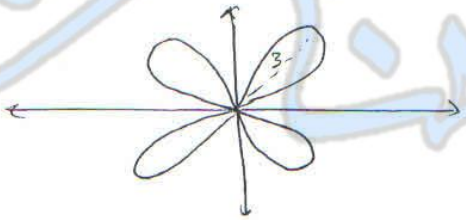
* Roses



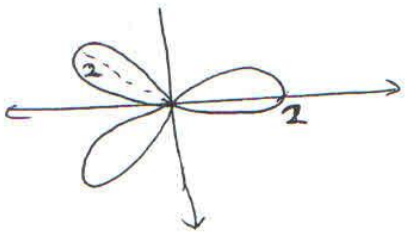
Examples:

Sketch

① $r = 3\sin 2\theta$

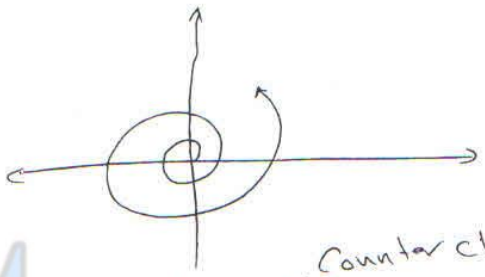


② $r = 2\cos 3\theta$



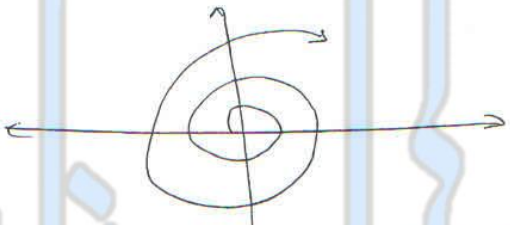
✱ Spirals

$r = a\theta$, $\theta > 0$



$\theta > 0 \rightarrow$ Counter clockwise

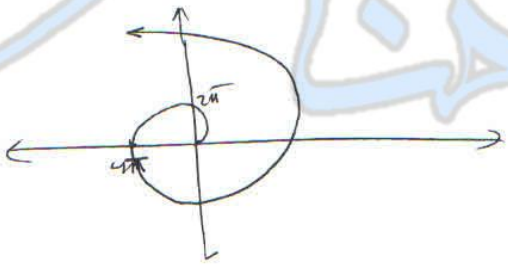
$r = a\theta$, $\theta < 0$



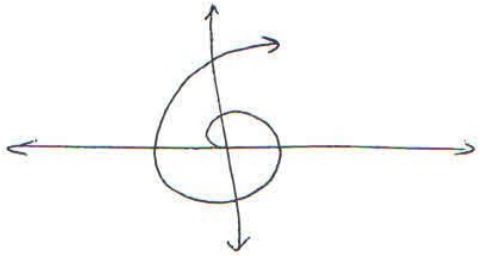
$\theta < 0 \rightarrow$ clockwise

Ex: Sketch

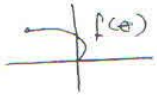
① $r = 4\theta$, $\theta \geq 0$



② $r = 4\theta$, $\theta < 0$



* Arc length in polar coordinates



$r = f(\theta)$, $\alpha \le \theta \le \beta$.

$$L = \int_{\alpha}^{\beta} \sqrt{(f(\theta))^2 + (f'(\theta))^2}\, d\theta = \int_{\alpha}^{\beta} \sqrt{r^2 + r'^2}\, d\theta$$

Ex: Find arc length of the spiral $r = e^{\theta}$ between $\theta = 0$ and $\theta = \pi$

$$L = \int_0^{\pi} \sqrt{e^{2\theta} + e^{2\theta}}\, d\theta = \sqrt{2} \int e^{\theta}\, d\theta = \sqrt{2}\, e^{\theta} \Big|_0^{\pi}$$

$$= \sqrt{2}\,(e^{\pi} - 1).$$

Ex: $r = 1 + \cos\theta$ $\quad 0 \le \theta \le 2\pi$

$$L = \int_0^{2\pi} \sqrt{(1+\cos\theta)^2 + (-\sin\theta)^2}\, d\theta = \int \sqrt{2 + 2\cos\theta}\, d\theta$$

$$= 2 \int \sqrt{1 + \cos\theta}\, d\theta = 2 \int \sqrt{\cos^2 \tfrac{1}{2}\theta}\, d\theta = 2 \int_0^{2\pi} |\cos \tfrac{1}{2}\theta|\, d\theta$$

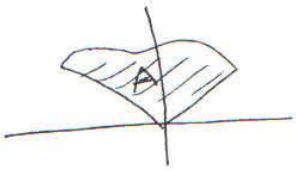
$$= 2 \left[\int_0^{\pi} \cos \tfrac{1}{2}\theta\, d\theta + - \int_{\pi}^{2\pi} \cos \tfrac{1}{2}\theta\, d\theta \right]$$



$$= 8.$$

* Area in Polar Coordinates

$r = f(\theta)$ cont. on $\alpha \le \theta \le \beta$.



$$A = \int_{\alpha}^{\beta} \frac{1}{2}(f(\theta))^2 \, d\theta = \int_{\alpha}^{\beta} \frac{1}{2} r^2 \, d\theta$$

Ex. Find area of the region in the 1st quadrant for $r = 1 - \cos\theta$

1st quadrant $\rightarrow 0 \le \theta \le \frac{\pi}{2}$.

$$A = \frac{1}{2} \int_{0}^{\frac{\pi}{2}} (1 - \cos\theta)^2 \, d\theta$$

$$= \frac{1}{2} \int 1 - 2\cos\theta + \cos^2\theta \, d\theta$$

$$= \frac{1}{2} \int 1 - 2\cos\theta + \frac{1 + \cos 2\theta}{2} \, d\theta$$

$$= \frac{1}{2} \left[\frac{3}{2}\theta - 2\sin\theta + \frac{\sin 2\theta}{4} \right|_{0}^{\frac{\pi}{2}}$$

$$= \frac{3\pi}{8} - 1$$